



金程教育®  
GOLDEN FUTURE

2025 FRM Part II  
百题巅峰班  
市场风险测量与管理

2025 年 4 月

## 1. Market Risk Measurement and Management

### 1.1. Estimating VaR and ES

#### 1.1.1. 重要知识点

##### 1.1.1.1. Normal VaR:

$$VaR_{\alpha} = -(\mu_{P/L} - z_{\alpha}\sigma_{P/L}) = \mu_{L/P} + z_{\alpha}\sigma_{L/P} = -(\mu_r - z_{\alpha}\sigma_r)P_{t-1}$$

##### 1.1.1.2. Lognormal VaR:

$$VaR_{\alpha} = (1 - e^{\mu_R - z_{\alpha}\sigma_R})P_{t-1}$$

##### 1.1.1.3. Expected Shortfall:

**Step 1:** slice the tail into a large number  $n$  of slices, each has the same probability mass.

**Step 2:** estimate the VaR associated with each slice.

**Step 3:** take the ES as the average of these VaRs.

#### 1.1.2. 基础题

**Q-1.** The VaR at a 95% confidence level is estimated to be 1.56 from historical simulation of 1,000 observations. Which of the following statements is most likely true?

- A. The parametric assumption of normal returns is correct
- B. The parametric assumption of lognormal returns is correct
- C. The historical distribution has fatter tails than a normal distribution.
- D. The historical distribution has thinner tails than a normal distribution.

**Q-2.** A risk manager is comparing the use of parametric and non-parametric approaches for calculating VaR and is concerned about some of the characteristics present in the loss data. Which of the following conditions would make non-parametric approaches the favored method to use?

- A. Scarcity of high magnitude loss event
- B. Skewness in the distribution
- C. Unusually high volatility during the data period
- D. Unusually low volatility during the data period

**Q-3.** A fund manager owns a portfolio of options on a non-dividend paying stock TUV. The portfolio is made up of 5,000 deep in-the-money call options on TUV and 20,000 deep out-of-the-money call options on TUV. The portfolio also contains 10,000 forward contracts on TUV. Currently, TUV is trading at USD 52. Assuming 252 trading days in a year and the volatility of TUV is 12% per year, which of the following amounts would be closest to the 1-day VaR of the portfolio at the 99% confidence level?

- A. USD 11,557
- B. USD 12,627

- C. USD 13,715
- D. USD 32,000

**Q-4.** A portfolio consists of options on Microsoft and AT&T. The options on Microsoft have a delta of 1000, and the options on AT&T have a delta of 20000. The Microsoft share price is \$120, and the AT&T share price is \$30. Assuming that the daily volatility of Microsoft is 2% and the daily volatility of AT&T is 1% and the correlation between the daily changes is 0.3, the 5-day 95% VaR is

- A. 26193
- B. 25193
- C. 27193
- D. 24193

**Q-5.** The annual mean and volatility of a portfolio are 10% and 40%, respectively. The current value of the portfolio is GBP 1,000,000. How does the 1-year 95% VaR that is calculated using a normal distribution assumption (normal VaR) compare with the 1-year 95% VaR that is calculated using the lognormal distribution assumption (lognormal VaR)?

- A. Lognormal VaR is greater than normal VaR by GBP 130,400
- B. Lognormal VaR is greater than normal VaR by GBP 17,590
- C. Lognormal VaR is less than normal VaR by GBP 130,400
- D. Lognormal VaR is less than normal VaR by GBP 17,590

**Q-6.** What is a key weakness of the value at risk (VaR) measure? VaR:

- A. Does not consider the severity of losses in the tail of the returns distribution.
- B. Is quite difficult to compute.
- C. Is subadditive.
- D. Has an insufficient amount of backtesting data.

**Q-7.** A wealth management firm has JPY 72 billion in assets. The portfolio manager computes the daily VaR at various confidence levels as follows:

| Confidence Level | VaR (USD)   |
|------------------|-------------|
| 95%              | 332,760,000 |
| 95.5%            | 336,292,500 |
| 96.0%            | 340,095,000 |
| 96.5%            | 350,332,500 |
| 97.0%            | 359,107,500 |

|       |             |
|-------|-------------|
| 97.5% | 367,882,500 |
| 98.0% | 378,412,500 |
| 98.5% | 392,452,500 |
| 99.0% | 410,880,000 |
| 99.5% | 439,252,500 |

What is the closest estimate of the daily ES at the 97.5% confidence level?

- A. JPY 398 million
- B. JPY 400 million
- C. JPY 405 million
- D. JPY 497 million

- Q-8.** A large commercial bank is using VaR as its main risk measurement tool. Expected shortfall (ES) is suggested as a better alternative to use during market turmoil. What should be understood regarding VaR and ES before modifying current practices?
- A. Despite being more complicated to calculate, ES is easier to backtest than VaR.
  - B. Relative to VaR, ES leads to more required economic capital for the same confidence level.
  - C. While VaR ensures that the estimate of portfolio risk is less than or equal to the sum of the risks of that portfolio's positions, ES does not.
  - D. Both VaR and ES account for the severity of losses beyond the confidence threshold.

- Q-9.** Value at risk (VaR) determines the maximum value we can lose for a given confidence level. For this reason, Kenneth Fulton is concerned that the VaR is not providing the magnitude of the actual loss. He has prepared the following table based on the assumption that returns are normally distributed and a corresponding  $n = 5$ . What is the expected shortfall using the information in the following table at the 95% confidence level?

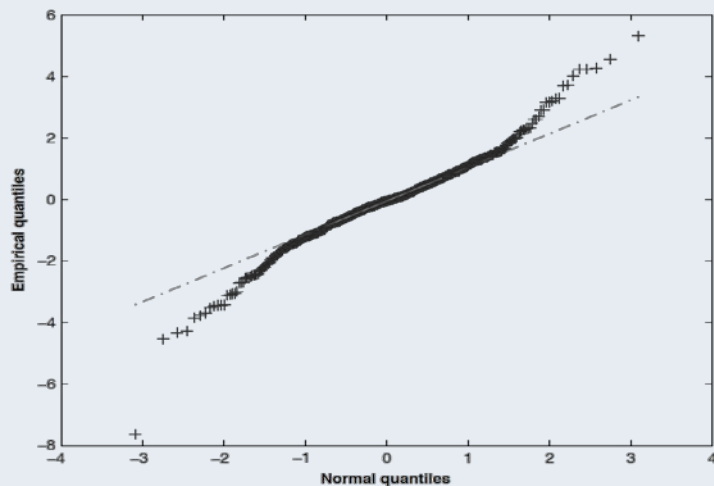
| Confidence level | VaR    |
|------------------|--------|
| 95%              | 1.6392 |
| 96%              | 1.7507 |
| 97%              | 1.8808 |
| 98%              | 2.0537 |
| 99%              | 2.3263 |

- A. 0.687
- B. 1.930
- C. 2.003
- D. 2.054

## 1.2. Quantile-Quantile Plots

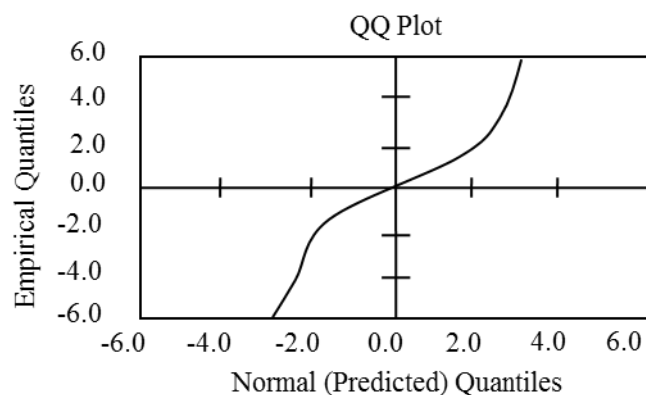
### 1.2.1. 重要知识点

**1.2.1.1. The quantile-quantile (QQ) plot** is a visual inspection of an empirical quantile relative to a hypothesized theoretical distribution. If the empirical distribution closely matches the theoretical distribution, the QQ plot would be linear.



### 1.2.2. 基础题

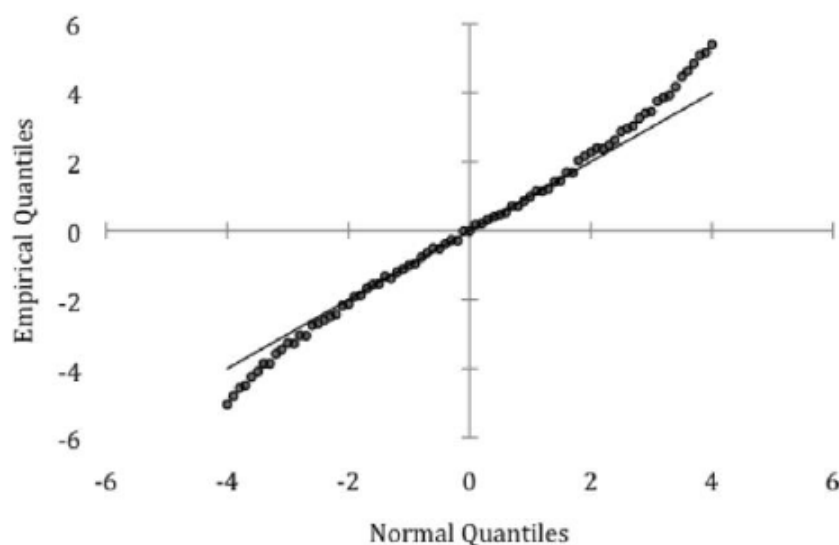
**Q-10.** Consider the following QQ plot:



Which is the most likely true statement about the QQ plot?

- A. The empirical distribution is actually parametric.
- B. The empirical distribution has positive skew.
- C. The empirical distribution has leptokurtosis (Excess kurtosis > 0)
- D. If we perform a linear transformation of location and scale, the distribution is approximately normal.

**Q-11.** An analyst is examining a sample of return data. As a first step, the analyst constructs a QQ plot of the data as shown below:



Based on an examination of the QQ plot, which of the following statements is correct?

- A. The returns are normally distributed.
- B. The return distribution has thin tails relative to the normal distribution.
- C. The return distribution is negatively skewed relative to the normal distribution.
- D. The return distribution has fat tails relative to the normal distribution.

**Q-12.** A risk analyst is studying a series of graphs plotting the empirical distribution of a portfolio's profit and loss quantiles against the quantiles of different specified reference distributions. The analyst plans to use these QQ plots to gain insights into the properties of the empirical distribution of the portfolio's profits and losses. Which of the following statements is correct regarding QQ plots?

- A. The tails of the empirical distribution are heavier than those of the reference distribution if the QQ plot has a steeper slope at its tails.
- B. QQ plots are used to smooth the empirical distribution by removing outliers through the use of quantiles.
- C. A distribution is considered a good reference distribution if data drawn from it produces a QQ plot that is non-linear in the central mass of the distribution.
- D. The primary purpose of a QQ plot is to measure the skewness and kurtosis of the empirical data.

**Q-13.** A risk analyst at an investment bank is examining how quantile estimators can be incorporated into the bank's risk measures. The analyst focuses on how estimators are constructed and how their precision and usefulness are determined. Which of the

following statements about quantile estimators is correct?

- A. Each quantile in the loss distribution must have an equal weight when used to create a coherent risk measure.
- B. The data and processes involved in estimating quantiles are different from those used to estimate coherent risk measures.
- C. QQ plots are a useful tool to evaluate the precision of a quantile estimator.
- D. Both the halving error and the standard error of a quantile estimator decrease as the number of quantiles used in the estimation process increases.

### 1.3. Non-Parametric Approach and Semi-Parametric Approach

#### 1.3.1. 重要知识点

**1.3.1.1. Basic historical simulation approach:** estimates VaR by means of ordered loss observations.

**1.3.1.2. Bootstrap historical simulation approach:** involves repeated sampling with replacement, the 5% VaR is recorded from each sample draw. The average of the VaRs from all the draws is the VaR estimate. Note: empirical analysis demonstrates that the bootstrapping technique consistently provides more precise estimates of coherent risk measures than historical simulation on raw data alone.

**1.3.1.3. Historical simulation using non-parametric density estimation:** a simple way is to draw in straight lines connecting the mid-points at the top of each histogram bar. Treating the area under the lines as a PDF enables us to estimate VaRs at any confidence level, regardless of the size of data set.

**1.3.1.4. Semi-Parametric approach:**

➤ **Age-weighted historical simulation:**

$$\omega_{(i)} = \frac{\lambda^{i-1}(1-\lambda)}{1-\lambda^n}$$

➤ **Volatility-weighted historical simulation:**

$$r_{t,i}^* = \left( \frac{\sigma_{T,i}}{\sigma_{t,i}} \right) r_{t,i}$$

➤ **Correlation-weighted historical simulation:** intuitively, the historical correlation matrix needs to be adjusted to the new information environment. This is accomplished, loosely speaking, by “multiplying” the historic returns by the revised correlation matrix to yield updated correlation-adjusted returns.

➤ **Filtered historical simulation:** combines the traditional historical simulation model with GARCH model.

#### 1.3.2. 基础题

- Q-14.** Johanna Roberto has collected a data set of 1,000 daily observations on equity returns. She is concerned about the appropriateness of using parametric techniques as the data appears skewed. Ultimately, she decides to use historical simulation and bootstrapping to estimate the 5% VaR. which of the following steps is most likely to be part of the estimation procedure?
- A. Filter the data to remove the obvious outliers.
  - B. Repeated sampling with replacement.
  - C. Identify the tail region from reordering the original data.
  - D. Apply a weighting procedure to reduce the impact of older data.
- Q-15.** Suppose you are using the volatility-weighted historical simulation approach to estimate value at risk (VaR) and expected shortfall (ES) for asset Y. The actual return for the asset 30 days ago was 1.5% with a daily volatility estimate of 1.0%. What is the volatility-adjusted return if the current daily volatility is 1.4%?
- A. 0.9%
  - B. 1.6%
  - C. 1.8%
  - D. 2.1%
- Q-16.** Jack has collected a large data set of daily market returns for three emerging markets and he want to compute the VaR. He is concerned about the non-normal skew in the data and is considering non-parametric estimation methods. Which of the following statements about age-weighted historical simulation approach is most accurate?
- A. The age-weighted procedure incorporate estimates from GARCH model.
  - B. If the decay factor in the model is close to 1, there is persistence within the data set.
  - C. When using this approach, the weight assigned on day  $i$  is equal to:  
$$\omega_{(i)} = \frac{\lambda^{i-1}(1 - \lambda)}{1 - \lambda^i}$$
  - D. The number of observation should at least exceed 250.
- Q-17.** If volatility (0) is the current (today's) volatility estimate and volatility (t) is the volatility estimate on a previous day (t), which best describes volatility-weighted historical simulation?
- A. First conduct typical historical simulation (HS) on return series. Then multiply VaR by volatility(0)/volatility(t)
  - B. First conduct typical historical simulation (HS) on return series. Then multiply VaR by volatility(t)/volatility(0)

- C. Each historical return (t) is replaced by:  $\text{return (t)} \times \text{volatility (0)} / \text{volatility (t)}$ . Then conduct typical historical simulation (HS) on adjusted return series.
- D. Each historical return (t) is replaced by:  $\text{return (t)} \times \text{volatility (t)} / \text{volatility (0)}$ . Then conduct typical historical simulation (HS) on adjusted return series.

**Q-18.** All of the following approaches improve the traditional historical simulation approach for estimating VaR except the:

- A. Volatility-weighted historical simulation.
- B. Age-weighted historical simulation.
- C. Market-weighted historical simulation.
- D. Correlation- weighted historical simulation.

**Q-19.** Which of the following statements about volatility-weighting is true?

- A. Historic returns are adjusted, and the VaR calculation is more complicated.
- B. Historic returns are adjusted, and the VaR calculation procedure is the same.
- C. Current period returns are adjusted, and VaR calculation is more complicated.
- D. Current period returns are adjusted, and VaR calculation is the same.

**Q-20.** All of the following items are generally considered advantages of non-parametric estimation methods except:

- A. Ability to accommodate skewed data.
- B. Availability of data.
- C. Use of historical data.
- D. Little or no reliance on covariance matrices.

**Q-21.** An analyst is evaluating how the bank applies historical simulation (HS) to estimate VaR and ES. The analyst focuses on the approaches used for weighting past return observations, including the age-weighted, volatility-weighted, correlation-weighted, and filtered HS approaches. Which of the following statements is correct regarding the given weighting approach?

- A. The age-weighted approach typically specifies that observations that occurred after the cutoff date are given an equal weight while observations that occurred before the cutoff date are given a weight of zero.
- B. The volatility-weighted approach adjusts historical returns in the sample by increasing them if the historical volatility forecast was higher than the current volatility forecast, and decreasing them if the historical volatility forecast was lower than the current volatility forecast.

- C. The correlation-weighted approach uses matrix multiplication to adjust historical portfolio returns so that these returns reflect current volatilities and current correlations.
- D. The filtered HS approach allows the historical returns data to be trimmed or customized based on day of the week, magnitude of return, or any other characteristic of the data.

## 1.4. Extreme-Value Theory

### 1.4.1. 重要知识点

#### 1.4.1.1. Generalized Extreme-Value Theory (Block Maxima)

- Consider a sample of size  $n$  drawn from  $F(x)$ , and let the maximum of this sample be  $M_n$ . If  $n$  is large, we can regard  $M_n$  as an extreme value.
- Under relatively general conditions, as  $n$  gets large the distribution of extremes (i.e.,  $M_n$ ) converges to generalized extreme-value (GEV) distribution.
- This distribution has three parameters.  $\mu$ , the location parameter of the limiting distribution, which is a measure of the central tendency of  $M_n$ ,  $\sigma$ , the scale parameter of the limiting distribution, which is a measure of the dispersion of  $M_n$ .  $\xi$ , the tail index, gives an indication of the shape (or heaviness) of the tail of the limiting distribution. If  $\xi > 0$ , the GEV becomes the Frechet distribution. This case is particularly useful for financial returns because they are typically heavy-tailed.

#### 1.4.1.2. Peaks-Over-Threshold (POT) Approach

- If  $x$  is a random i.i.d. loss with distribution function  $F(x)$ , and  $u$  is a threshold value of  $x$ , we can define the distribution of excess losses over threshold  $u$ .
- The distribution of  $x$  itself can be any of the commonly used distributions and will usually be unknown to us. However, as  $u$  gets large, the distribution  $F_u(x)$  converges to a generalized Pareto distribution.
- Two parameters: a positive scale parameter,  $\beta$ , and a shape or tail index parameter,  $\xi$ . This latter parameter is the same as the tail index encountered already with GEV theory.

#### 1.4.1.3. Differences

- Both are different manifestations of the same underlying EV theory.
- POT model exceedances over a high threshold while GEV theory model the maxima of a large sample.
- POT require fewer parameters.
- The block maxima approach can involve some loss of useful data, because some blocks might have more than one extreme in them.
- POT requires us to grapple with the problem of choosing the threshold.

#### 1.4.1.4. ES and VaR Estimation with POT

$$VaR_{\alpha} = \begin{cases} u + \frac{\beta}{\xi} \left\{ \left[ \frac{n}{N_u} (1 - \alpha) \right]^{-\xi} - 1 \right\} & \text{if } \xi \neq 0 \\ u - \beta \ln \left[ \frac{n}{N_u} (1 - \alpha) \right] & \text{if } \xi = 0 \end{cases}$$

$$ES_{\alpha} = \begin{cases} \frac{VaR_{\alpha}}{1 - \xi} + \frac{\beta - \xi u}{1 - \xi} & \text{if } \xi \neq 0 \\ VaR_{\alpha} + \beta & \text{if } \xi = 0 \end{cases}$$

➤ Where  $u$  is the threshold,  $N_u$  is the number of observations in excess of the threshold value,  $\alpha$  is the confidence level.

#### 1.4.2. 基础题

**Q-22.** The generalized extreme value (GEV) generally requires:

- A. Fewer estimated parameters than the POT approach and does not share any parameters with the POT approach.
- B. Fewer estimated parameters than the POT approach and shares one parameter with the POT.
- C. More estimated parameters than the POT approach and shares one parameter with the POT.
- D. More estimated parameters than the POT approach and does not share any parameters with the POT approach.

**Q-23.** In setting the threshold in the POT approach, which of the following statements is the most accurate? Setting the threshold relatively high makes the model:

- A. Less applicable but increases the number of observations in the modeling procedure.
- B. Less applicable and decreases the number of observations in the modeling procedure.
- C. More applicable but decreases the number of observations in the modeling procedure.
- D. More applicable but increases the number of observations in the modeling procedure.

**Q-24.** An investment bank with an active position in commodity futures is using the peaks-over-threshold (POT) methodology for estimating VaR and ES at the 99% confidence level. The bank's risk managers have set a threshold level to evaluate excess losses. The choice of the threshold, they argue, is suitable and consistent with the finding that 5.00% of the observations are in excess of the threshold value. The risk managers have concluded that the position's VaR using the POT measure is 4.45%. The VaR estimate is computed from the following parameters and the managers' empirical analysis is based upon the generalized Pareto distribution assumption for the excess losses.

| Parameter              | Symbol | Value |
|------------------------|--------|-------|
| Loss threshold         | $u$    | 3     |
| Number of observations | $N$    | 740   |

|                                              |         |      |
|----------------------------------------------|---------|------|
| Number of observations that exceed threshold | n       | 37   |
| Scale                                        | $\beta$ | 0.75 |
| Shape (tail index)                           | $\xi$   | 0.22 |

Given the VaR value and the parameter assumptions, which of the following is correct?

- A. Keeping all other parameters constant, increasing the value of the tail index lowers both the ES and the VaR.
- B. Keeping all other parameters constant, increasing the loss threshold level increases both the ES and the VaR.
- C. The value of ES is 4.57%
- D. The value of ES is 5.71%

**Q-25.** A CRO is concerned that existing internal risk models of a firm, which are governed mainly by the central limit theorem, are not adequate in addressing potential random extreme losses of the firm. The CRO then recommends the use of extreme value theory (EVT). When applying EVT and examining distributions of losses exceeding a threshold value, which of the following is correct?

- A. As the threshold value is increased, the distribution of losses over a fixed threshold value converges to a generalized Pareto distribution.
- B. If the tail parameter value of the generalized extreme-value (GEV) distribution goes to infinity, then the GEV essentially becomes a normal distribution.
- C. To apply EVT, the underlying loss distribution must be either normal or lognormal.
- D. The number of exceedances decreases as the threshold value decreases, which causes the reliability of the parameter estimates to increase.

**Q-26.** A risk analyst at an investment bank is evaluating the bank's application of extreme value theory (EVT) in managing financial risks. The analyst compares different methods of estimating extreme value (EV) parameters for the bank's operational loss distribution and examines the mechanics of each method, as well as their advantages and disadvantages. Which of the following statements regarding a method used for estimating EV parameters is correct?

- A. The regression method uses an ordered sample of losses to obtain least squares estimates of the EV parameters.
- B. The maximum-likelihood method uses the average of a random number of the most extreme observations to estimate EV parameters.
- C. The main challenge associated with the moment-based method is choosing the number of observations that minimizes the mean-squared-error loss function.
- D. A drawback of the semi-parametric estimation method is that the Hill estimator is neither

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consistent nor asymptotically normal.

## 1.5. Exceedance-Based Backtesting of VaR

### 1.5.1. 重要知识点

#### 1.5.1.1. Binomial Distribution Test

- $H_0$ : accurate model  $H_a$ : inaccurate model
- Test statistic:

$$z = \frac{x - Tp}{\sqrt{Tp(1-p)}} \xrightarrow{d} N(0, 1)$$

p: the probability of exception,  $p = 1 - c$

x: the number of exceptions

T: the number of samples

#### 1.5.1.2. Kupiec's Proportion of Failures Test

- $H_0$ : accurate model  $H_a$ : inaccurate model
- Test statistic:

$$LR_{POF} = 2 \left\{ \ln \left[ \left( \frac{x}{T} \right)^x \left( 1 - \frac{x}{T} \right)^{T-x} \right] - \ln[(p)^x (1-p)^{T-x}] \right\} \xrightarrow{d} \chi_1^2$$

- If  $LR > 3.84$ , we would reject the hypothesis that the model is correct.

#### 1.5.1.3. Basel Committee's Traffic Light Test

| Basel Penalty Zones |                      |               |
|---------------------|----------------------|---------------|
| Zone                | Number of Exceptions | Multiplier(k) |
| Green               | 0 to 4               | 3.00          |
| Yellow              | 5                    | 3.40          |
|                     | 6                    | 3.50          |
|                     | 7                    | 3.65          |
|                     | 8                    | 3.75          |
|                     | 9                    | 3.85          |
| Red                 | 10 or more           | 4.00          |

#### 1.5.1.4. Christoffersen's Test

- **Unconditional coverage:** the number of exceptions which may be observed in a given period is in line with the VaR confidence level.
- **Independence:** the observed exceptions should be independent from one another, and each observed exception should not be informative of future exceptions.
- The ultimate objective of the Christoffersen's test is to test jointly for independence and unconditional coverage.

**1.5.2. 基础题**

**Q-27.** Which of the following statements regarding verification of a VaR model by examining its failure rates is false?

- I. The frequency of exceptions should correspond to the confidence level used for the model.
- II. According to Kupiec (1995), we should reject the hypothesis that the model is correct if the  $LR > 3.84$ .
- III. Backtesting VaR models with lower confidence levels is difficult because the number of exceptions is not high enough to provide meaningful information.
- IV. The range for the number of exceptions must strike a balance between the chances of rejecting an accurate model (a type 1 error) and the chance of accepting an inaccurate model (a type 2 error).

- A. I and IV
- B. II only
- C. III only
- D. II and IV

**Q-28.** A risk manager is concerned that his value at risk (VaR) model is complicated by intraday changes as well as profit and loss factors. Which of the following backtesting techniques will most likely mitigate these issues? Backtest the VaR model using:

- A. shorter time periods, such as one year as opposed to five years.
- B. daily holding period returns.
- C. a higher confidence level, such as 99%.
- D. a lower confidence level, such as 95%.

**Q-29.** Basel II requires a backtest at a 99% confidence level of a bank's internal value at risk (VaR) model (IMA). Assume the bank's ten-day 99% VaR is \$1 million (minimum of 99% is hard-wired per Basel). The null hypothesis is: the VaR model is accurate. Out of 1,000 observations, 25 exceptions are observed (we saw the actual loss exceed the VaR 25 out of 1000 observations).

- A. We will probably call the VaR model good (accurate) but we risk a Type I error.
- B. We will probably call the VaR model good (accurate) but we risk a Type II error.
- C. We will probably call the model bad (inaccurate) but we risk a Type I error.
- D. We will probably call the model bad (inaccurate) but we risk a Type II error.

**Q-30.** A bank conducted a backtest of its 95% daily value at risk (VaR) and observed 19 exceptions - i.e., the number of days where the daily P&L loss exceeded the VaR - over the last year

which included 250 trading days ( $T = 250$ ). If we use the normal distribution to approximate the binomial for purposes of model verification, what is our accept/reject opinion of the model under a 90% two-tailed test?

- A. Accept with a test statistic of 1.25
- B. Accept with a test statistic of 1.89
- C. Reject with a test statistic of 1.25
- D. Reject with a test statistic of 1.89

**Q-31.** A newly hired risk analyst is backtesting a firm's VaR model. Previously, the firm calculated a 1-day VaR at the 95% confidence level. Following the Basel framework, the risk analyst is recommending that the firm switch to a 99% VaR confidence level. Which of the following statements concerning this switch is correct?

- A. The decision to accept or reject a VaR model based on backtesting results at the two-tailed 95% confidence level is less reliable with a 99% VaR model than with a 95% VaR model.
- B. The 95% VaR model is less likely to be incorrectly rejected using backtesting than the 99% VaR model.
- C. When backtesting using a two-tailed 90% confidence level test, there is a smaller probability of incorrectly rejecting a 95% VaR model than a 99% VaR model.
- D. Using a 99% VaR model will lower the probability of committing both type 1 and type 2 errors.

**Q-32.** Based on Basel II rules for backtesting, a penalty is given to banks that have more than four exceptions to their 1-day 99%VaR over the course of 250 trading days. The supervisor gives these penalties based on four criteria. Which of the following causes of exceptions is most likely to lead to a penalty?

- A. The bank increases its intraday trading activity.
- B. A large move in interest rates was combined with a small move in correlations.
- C. The bank's model calculates interest rate risk based on the median duration of the bonds in the portfolio.
- D. A sudden market crisis in an emerging market leads to losses in the equity positions in that country.

**Q-33.** A risk manager is analyzing a 1-day 99% VaR model. Assuming 225 days in a year, what is the maximum number of daily losses exceeding the 1-day 99% VaR that is acceptable in a 1-year backtest to conclude, at a 95% confidence level, that the model is calibrated correctly?

- A. 3
- B. 5
- C. 8
- D. 10

**Q-34.** An analyst at bank LKS has been asked to validate the bank's VaR model through backtesting. The analyst uses two sets of returns data to generate results of predicted and actual losses that can be compared in the validation process. Which of the following correctly describes the two most appropriate sets of returns data to use in backtesting?

- A. The actual returns, which correspond to the total return on the bank's trading portfolio, and the hypothetical returns, which represent the returns obtained from freezing the starting positions in the bank's trading portfolio
- B. The cleaned returns, which are the actual returns minus any profit and loss from intraday trades, and the actual returns, which correspond to the total returns on the bank's trading portfolio
- C. The hypothetical returns, which represent the returns obtained from freezing the starting positions in the bank's trading portfolio, and the cleaned returns, which are the actual returns minus any profit and loss from intraday trades
- D. The trading returns, which are the actual returns minus any fees and commissions, and the hypothetical returns, which represent the actual returns obtained from freezing the starting positions in the bank's trading portfolio.

**Q-35.** A regulatory analyst is reviewing the Basel Committee rules for backtesting VaR models. The analyst notes that under the Basel framework, a penalty can be given to banks that have more than four exceptions to their 1-day 99% VaR over the last 250 trading days. Which of the following scenarios is most likely to result in a penalty?

- A. A large move in interest rates occurs in conjunction with a small move in correlations.
- B. The bank's model calculates interest rate risk based on the median duration of the bonds in the portfolio.
- C. A sudden market crisis in an emerging market, which leads to losses in the equity positions in that country.
- D. A sudden devastating earthquake that causes major losses in the bank's key area of operation.

**Q-36.** An analyst at a bank is asked to evaluate the efficacy of the VaR model that the bank used over the last year. While evaluating the model, the analyst finds that the bank experienced more VaR exceptions than were forecast by the bank's VaR model and examines both the

reason this occurred and its potential impact on the bank. Which of the following is correct regarding the analyst's examination of the situation?

- A. The confidence level of the backtest performed on the VaR model was set too low.
- B. The model is most likely overestimating the market risk faced by the bank .
- C. The risk-taking units of the bank will likely be allocated less capital than they should be.
- D. The bank's regulator will impose a financial penalty on the bank.

**Q-37.** A model validation team at a bank is backtesting the bank's VaR model. In preparation for the backtest, one of the team members expresses a concern that the validation process could result in the team committing a Type I error or a Type II error and discusses the characteristics of these errors with the team. Which of the following is correct regarding Type I and Type II errors?

- A. The probability of committing a Type II error decreases when the sample size increases and the level of significance is held constant.
- B. A backtest is considered statistically powerful if it minimizes the probability of committing a Type II error regardless of the probability of committing a Type I error.
- C. A Type I error is committed when an incorrectly specified model is accepted.
- D. A Type II error is committed when a correctly specified model is rejected.

## 1.6. PIT-Based Backtesting of VaR

### 1.6.1. 重要知识点

**1.6.1.1. Probability Integral Transform (PIT):** the CDF of a random variable is itself a random variable and follows the standard uniform distribution.

**1.6.1.2. Applying the PIT in Backtesting:** if the risk model is perfectly accurate, the series of  $p_{t+1} = F_t(P/L_{t+1})$  should be iid  $U(0, 1)$ . Operationally:

- The series of  $p_{t+1}$  are  $U(0, 1)$  distributed corresponds to that the model satisfies the unconditional coverage property.
- Further satisfying the independence property corresponds to  $p_{t+1} \sim \text{iid } U(0, 1)$ .

#### 1.6.1.3. Potential Model Misspecifications

- **Conservative misspecification:** the model distribution is too wide, and P&L observed is small, clustering in the middle of the distribution.
- **Aggressive misspecification:** the model distribution is too narrow and we often see realized P&L in the tails of the distribution.

#### 1.6.1.4. Testing the PITs Uniformity

- **Kolmogorov-Smirnov (KS) test:** the KS test is a widely used nonparametric test that evaluates the goodness-of-fit by comparing the empirical distribution

function of a sample with some reference distribution function based on the supremum (greatest distance). Its test statistic is equal to 0 under the null that the data follows the reference distribution. It is noted that a limitation of this test is its lack of sensitivity at the extremes of the distribution.

- **Anderson-Darling (AD) test:** the AD test is a modification of the KS test that corrects for the limitation discussed above, which puts more weight on the tails of the distribution. As with the KS test, the AD test statistic is equal to zero under the null that the data follows the reference distribution.
- **Cramér-von Mises test:** the Cramér-von Mises test is another variation of the KS test that relies on the mean squared deviation of the distribution. As with the KS test, the Cramér-von Mises test statistic is equal to zero under the null that the data follows the reference distribution.

### 1.6.2. 基础题

- Q-38.** Any backtesting of a VaR model could be reduced to the problem of determining whether the sequence of exceptions satisfies two properties, namely the unconditional coverage property and the independence property. Which of the following statements correctly links these two properties to the backtesting based on probability integral transform (PIT)?
- A. The unconditional coverage property requires the consistency between the number of observed exceptions and the VaR confidence level, which corresponds to a standard normally distributed PIT series.
  - B. The unconditional coverage property restricts the number of observed exceptions at a determined significance level, which corresponds to a standard uniformly distributed PIT series.
  - C. The independence property requires that each observed exception is informative of future exceptions, which corresponds to an iid standard uniformly distributed PIT series.
  - D. The independence property requires that the observed exceptions should be independent from one another, which corresponds to an iid standard normally distributed PIT series.
- Q-39.** A US bank is required to submit PIT-based backtesting information for each sub-portfolio, for each business day, on an ongoing basis. Which of the following statements correctly describe the derivation of PITs performed by the bank?
- A. The bank collects the daily realization of the portfolio Profit/Loss and calculate the risk model's probability of observing a realization below the actual Profit/Loss.
  - B. The bank collects the daily realization of the portfolio Profit/Loss and calculate the risk model's probability of observing a realization equal to the actual Profit/Loss.

- C. The bank collects the daily realization of the portfolio Profit/Loss and calculate the risk model's probability of observing a realization above the actual Profit/Loss.
- D. The bank collects the portfolio Profit/Loss predicted by the risk model and calculate the probability of observing a realization below the predicted Profit/Loss based on the empirical distribution.

**Q-40.** The degree to which actual distribution of PITs differs from uniform may provide information regarding the model's ability to accurately and reliably assess risk of a given portfolio, and the deviations indicate that the model is likely to be either conservatively or aggressively misspecified. Which of the following statements correctly describes the potential model misspecification?

- A. A model is conservatively misspecified when the model distribution is too wide, and we often see realized Profit/Loss in the tails of the distribution.
- B. A model is conservatively misspecified when the model distribution is too narrow, and the observed Profit/Loss clusters in the middle of the distribution.
- C. A model is aggressively misspecified when the model distribution is too wide, and the observed Profit/Loss clusters in the middle of the distribution.
- D. A model is aggressively misspecified when the model distribution is too narrow, and we often see realized Profit/Loss in the tails of the distribution.

**Q-41.** Cindy, a post-graduate student in risk management, is learning how to use various goodness-of-fit tests to assess the uniformity of the distribution of PITs. Which of the following statements correctly describes these tests?

- A. One limitation of the Kolmogorov-Smirnov test is its lack of sensitivity at the center of the distribution, so that it is ineffective in detecting differences in central masses of the two distributions.
- B. The Cramer-von Mises test is a variation of the Kolmogorov-Smirnov test that relies on the mean absolute deviation of the distribution.
- C. In the context of VaR backtesting, the null hypothesis of Anderson-Darling test is that the empirical CDF is not cumulative standard uniform.
- D. When the null hypothesis is true, the test statistic of all three goodness-of-fit tests is equal to zero.

## **1.7. Validating VaR Models**

### **1.7.1. 重要知识点**

#### **1.7.1.1. Assess the Conceptual Soundness of the Model:**

- **Core decision in building a model:** whether the model is suitable for the purpose it is developed for. For VaR models, the basic purpose is to help the firm manage the risk of positions. In doing so, we require that the change in the firm's positions is reflected in VaR.

- **Sensitivity analysis framework:**

$$VaR(V_{PT}) = \sum_{i \in P} component_i VaR(V_{PT})$$

$$component_i VaR(V_{PT}) = VaR(V_{PT}) \omega_i \beta_i$$

$\omega_i$ : the share of the position in the portfolio.

$\beta_i$ : the sensitivity of the position value to the overall portfolio value, which can be estimated from  $\Delta V_{it}(\text{or } r_{it}) = \alpha_i + \beta_i \times \Delta V_{Pt}(\text{or } r_{Pt}) + \varepsilon_t$ .

#### 1.7.1.2. Confidence Interval for VaR:

- **Based on the asymptotic standard error of a quantile:**

$$[\widehat{VaR}_{PT} - z_{\alpha} SE(VaR_{PT}), \widehat{VaR}_{PT} + z_{\alpha} SE(VaR_{PT})]$$

- **Based on order statistics:** the  $r$ th order statistic is the  $r$ th lowest in a sample of  $n$  observations, which is itself a random variable and has its distribution function. VaR can be considered as an order statistic in the loss variables, and we can use the distribution function of order statistics to construct the VaR distribution function and calculate the confidence interval.
- **Based on bootstrap:** by constructing  $n$  bootstrapped resamples and obtain  $n$  VaR estimates, we can estimate a confidence interval for VaR estimator.

#### 1.7.1.3. Benchmarking VaR Models: the formal comparison between a bank's risk model and an alternative model.

### 1.7.2. 基础题

**Q-42.** Model validation requires a review of conceptual soundness of the model, which requires the validator to carefully examine the assumptions, techniques and data used by the model. Which of the following statements is correct regarding the important considerations in assessing the VaR model's conceptual soundness?

- The core of model validation is to examine whether the model is suitable for the purpose it is developed for, and for VaR models the basic purpose is to enhance capital calculation methods.
- The VaR model calculated based on actual Profit/Loss history would accurately, but not timely, reflect the change in portfolio risk once the positions are changed.
- Incorporating positional information and recomputing the pseudo history of the change in the portfolio whenever the portfolio changes will produce a risk measure that can better capture the portfolio risk dynamic.

- D. The conceptual soundness of the VaR model can only be considered using qualitative check, and the main consideration is all about how the model will be used.

**Q-43.** Bruce, FRM, is testing the sensitivity of the VaR model to changes in positions. Which of the following statements is correct regarding the VaR model's sensitivity analysis?

- A. The sensitivity of VaR to the position is dependent on the share of the position in the portfolio and the sensitivity of the overall portfolio value to the position value.
- B. Data scarcity is negligible in conducting the sensitivity analysis, because the regression used in testing the sensitivity is based on simulated value changes of the position and portfolio.
- C. The sensitivity analysis can help the financial institution to examine some choices it has made regarding the valuation of a particular position.
- D. The portfolio composition of a large trading organization typically changes slowly, which does not form a challenge of performing the sensitivity analysis.

**Q-44.** It is natural to ask about the accuracy of the VaR estimate. In order to evaluate this, we can compute the confidence intervals around the estimate whereby the true VaR should be within the interval a high percentage of the time. Which of the following statements correctly describes the approaches used to compute the confidence intervals?

- A. It is desirable to base the confidence interval on the single point of the quantile estimate, and this view supports the use of asymptotic standard error of a quantile to construct the interval.
- B. The use of order statistics starts with the pseudo history of portfolio value changes (denoted by  $\Delta V$ ), and reorders them from highest to lowest so that  $\Delta V_{(1)} \geq \Delta V_{(2)} \geq \dots \geq \Delta V_{(T)}$ .
- C. The advantage of using order statistics over using bootstrap procedure is that the order statistics approach does not assume independence of the changes in portfolio value over time.
- D. Under the bootstrap procedure, the distribution of VaR is the sampling distribution, and the confidence interval can be calculated from the ordered VaR values.

**Q-45.** Benchmarking refers to the action of formally comparing a bank's model with an alternative model. Which of the following statements correctly describe the common practice in benchmarking?

- A. The most common practice is to plot the two VaR models over time, and this method is considerably informative in comparing the models.
- B. The loss function used in comparing VaR models favors conservatism in the estimate of

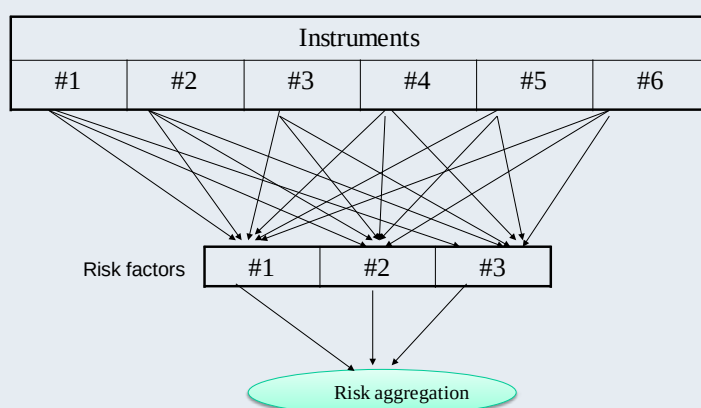
VaR since only exceedances are penalized.

- C. The issue that trading portfolios change frequently with non-iid errors can be overcome by estimating a GARCH (1, 1) VaR model on the Profit/Loss from trading by the bank.
- D. The sign test is a test of the median of the distribution of the loss differential between two competing models, and rejecting the null would indicate that the two models are equally good.

## 1.8. VaR Mapping

### 1.8.1. 重要知识点

#### 1.8.1.1. Framework



#### 1.8.1.2. Mapping Stock Portfolios

- Single index model:

$$r_i = \alpha_i + \beta_i r_M + \varepsilon_i$$

- Variance of portfolio return:

$$\begin{aligned}\sigma_P^2 &= \left( \sum_{i=1}^N w_i \beta_i \right)^2 \sigma_M^2 + \sum_{i=1}^N w_i^2 \sigma_{\varepsilon_i}^2 \\ &= \beta_P^2 \sigma_M^2 + \sum_{i=1}^N w_i^2 \sigma_{\varepsilon_i}^2 \\ &= \text{General Risk} + \text{Specific Risk}\end{aligned}$$

- Portfolio VaR (assuming zero specific risk):

$$\begin{aligned}VaR_P &= z_\alpha \sigma_P V_P \\ &= z_\alpha \beta_P \sigma_M V_P\end{aligned}$$

#### 1.8.1.3. Mapping Fixed Income Portfolios

- **Principal mapping:** Only the risk associated with the return of principal at the maturity of the bond is mapped. Principal mapping includes only the risk of repayment of the principal amounts.

- **Duration mapping:** The risk of the bond is mapped to a zero-coupon bond of the same duration. Duration mapping uses the duration of the portfolio to calculate the VaR.
- **Cash flow mapping:** The risk of the bond is decomposed into the risk of each of the bonds' cash flows. Cash flow mapping is the most precise method because we map the present value of the cash flows (face amount discounted at the spot rate for that maturity) onto the risk factors for zeros of the same maturities and include the inter-maturity correlations.

#### 1.8.1.4. Mapping Derivatives

- Long currency forward contract = long foreign currency spot + long foreign currency bill + short U.S. dollar bill
- Long 6×12 FRA = long 6-month bill + short 12-month bill
- Long call option = long  $N(d_1)$  asset + short  $N(d_2)$  bill
- Long put option = long  $N(-d_2)$  bill + short  $N(-d_1)$  asset
- Linear approximations may be acceptable for options with long maturities when the risk horizon is short.

#### 1.8.2. 基础题

**Q-46.** Alice is managing a diversified portfolio consisting of three independent assets: A, B, and C. The portfolio is designed to track a benchmark index, but it also has some specific risk due to the unique characteristics of the selected assets and their weights in the portfolio. The following information is provided:

| Asset | Beta | Weight in the portfolio |
|-------|------|-------------------------|
| A     | 1.2  | 0.5                     |
| B     | 0.9  | 0.2                     |
| C     | 0.8  | 0.3                     |

Portfolio variance ( $\sigma_p^2$ ) = 0.0205

Benchmark index variance ( $\sigma_m^2$ ) = 0.0225

The specific risk of the portfolio (also known as idiosyncratic risk) is the portion of the portfolio's variance that cannot be explained by the benchmark index. What is the portfolio specific risk in variance?

- A. 0.0168
- B. 0.0185
- C. 0.0192
- D. 0.0200

**Q-47.** An analyst is using the delta-normal method to determine the VaR of a fixed income

portfolio. The portfolio contains a long position in 1-year bonds with a \$1 million face value and a 6% coupon that is paid semi-annually. The interest rates on six-month and twelve-month maturity zero-coupon bonds are, respectively, 2% and 2.5%. Mapping the long position to standard positions in the six-month and twelve-month zeros, respectively, provides which of the following mapped positions?

- A. \$30,000 and 1,030,000
- B. \$29,500 and 975,610
- C. \$29,703 and 1,004,878
- D. \$30,300 and 1,035,000

**Q-48.** Which of the following can be considered a general risk factor?

- I. Exchange rate
  - II. Mortgage-backed securities
  - III. Zero-coupon bond
  - IV. Interest rate
- A. I only
  - B. II and III
  - C. III only
  - D. I and IV

**Q-49.** Delta-normal VaR will provide accurate estimates for option contracts when:

- A. Deltas are stable
- B. Options are at the money
- C. The correlation matrix is available
- D. The delta-normal method can never be used for option contracts

**Q-50.** Under these assumptions - in particular: a flat yield curve and constant yield volatility of 1.0% - why can we expect cash flow mapping to produce a lower diversified VaR than either duration and principal mapping?

- A. The risk measures are non-linear.
- B. Due to imperfect correlations between pairwise risk factors.
- C. Fewer total cash flows will be mapped.
- D. We cannot expect a lower diversified VaR.

**Q-51.** In fixed income portfolio mapping, when the risk factors have been selected, which of the following mapping approaches requires that one risk factor be chosen that corresponds to average portfolio maturity?

- A. Principal mapping
- B. Duration mapping
- C. Convexity mapping
- D. Cash flow mapping

**Q-52.** A portfolio manager is mapping a fixed-income portfolio into exposures on selected risk factors. The manager is analyzing the comparable mechanics and risk measurement outputs of principal mapping, duration mapping, and cash-flow mapping. Which of the following is correct?

- A. Cash-flow mapping groups cash flows into buckets based on their size.
- B. Cash-flow mapping uses the average rates in each risk group as a discount factor.
- C. Principal mapping incorporates correlations among zero-coupon bonds.
- D. Duration mapping replaces the portfolio with a zero-coupon bond with maturity equal to the duration of the portfolio.

**Q-53.** Which of these statements regarding risk factor mapping approaches is/are correct?

- I. Under the cash flow mapping approach, only the risk associated with the average maturity of a fixed-income portfolio is mapped.
  - II. Cash flow mapping is the least precise method of risk mapping for a fixed-income portfolio.
  - III. Under the duration mapping approach, the risk of a bond is mapped to a zero-coupon bond of the same duration.
  - IV. Using more risk factors generally leads to better risk measurement but also requires more time to be devoted to the modeling process and risk computation.
- A. I and II
  - B. I, III, and IV
  - C. III and IV
  - D. IV only

**Q-54.** Computing VaR on a portfolio containing a very large number of positions can be simplified by mapping these positions to a smaller number of elementary risk factors. Which of the following mappings would be adequate?

- A. USD/EUR forward contracts are mapped on the USD/JPY spot exchange rate.
- B. Each position in a corporate bond portfolio is mapped on the bond with the closest maturity among a set of government bonds.
- C. Government bonds paying regular coupons are mapped on zero-coupon government bonds.

- D. A position in the stock market index is mapped on a position in a stock within that index.

**Q-55.** A portfolio manager is mapping a fixed-income portfolio into exposures on selected risk factors. The manager is analyzing the comparable mechanics and risk measurement outputs of principal mapping, duration mapping, and cash-flow mapping that correspond to the average portfolio maturity. Which of the following is correct?

- A. Principal mapping considers coupon and principal payments, and the portfolio VaR using principal mapping is greater than the portfolio VaR using cash-flow mapping.
- B. Duration mapping does not consider intermediate cash flows and the portfolio VaR using such method is less than the portfolio VaR using principal mapping.
- C. Cash-flow mapping considers the timing of the redemption cash flow payments only, and the portfolio VaR using cash flow mapping is less than the portfolio VaR using duration mapping.
- D. Cash-flow mapping considers the present values of cash flows grouped into maturity buckets, and the undiversified portfolio VaR using cash-flow mapping is greater than the portfolio VaR using principal mapping.

**Q-56.** An analyst at a fixed-income investment company is evaluating different ways the company uses to estimate the VaR of its corporate bond portfolios. The portfolios consist of a large number of bonds with a wide range of maturities. The analyst examines the possibility of using a mapping approach to simplify the estimation process. Which of the following statements would the analyst be correct to make regarding the approaches to mapping fixed-income portfolios?

- A. The VaR estimated using the principal mapping approach understates the true risk of a portfolio since it ignores coupon payments and any risk associated with them.
- B. The VaR estimated using the duration mapping approach replaces the portfolio with a zero-coupon bond whose maturity equals the duration of the portfolio.
- C. The VaR estimated using the principal mapping approach differs from the undiversified VaR estimated using the duration mapping approach due to an adjustment made for correlations.
- D. The VaR estimated using the cash-flow mapping approach is less accurate than the VaR estimated using the duration mapping approach since it does not account for the timing of cash flows.

**Q-57.** A fund manager owns a portfolio of options on TUV, a non-dividend paying stock. The portfolio is made up of 5,000 deep in-the-money call options on TUV and 20,000 deep out-

of-the-money call options on TUV. The portfolio also contains 10,000 forward contracts on TUV. Currently, TUV is trading at USD 52. Assuming 252 trading days in a year, the volatility of TUV is 12% per year, and that each of the option and forward contracts is on one share of TUV, which of the following amounts would be closest to the 1-day 99% VaR of the portfolio?

- A. USD 11,557
- B. USD 12,627
- C. USD 13,715
- D. USD 32,000

## 1.9. Correlations and Copulas

### 1.9.1. 重要知识点

**1.9.1.1. Correlation Swap:** Paying a fixed rate in a correlation swap is also called buying correlation. The payoff =  $NP \times (\text{realized } \rho - \text{fixed } \rho)$ .

$$\rho_{\text{realized}} = \frac{2}{n^2 - n} \sum_{i>j} \rho_{i,j}$$

There is a positive relationship between correlation and volatility.

### 1.9.1.2. Copula:

$$C_{GD}[Q_i(t), Q_n(t)] = M_n [N^{-1}(Q_i(t)), N^{-1}(Q_n(t)); \rho_M]$$

$Q_i(t)$ : cumulative default probability of asset  $i$  at time  $t$ .

$C_{GD}$ : Gaussian default time copula.

$M_n$ : the joint,  $n$ -variate cumulative standard normal distribution.

$\rho_M$ : the  $n \times n$  symmetric, positive-definite correlation matrix of the  $n$ -variate normal distribution  $M_n$ .

### 1.9.1.3. Mean reversion

Mean reversion is present if there is a negative relationship between the change of a variable,  $\rho_t - \rho_{t-1}$ , and the variable  $\rho_{t-1}$ .

$$\begin{aligned} \rho_t - \rho_{t-1} &= \kappa(\mu_\rho - \rho_{t-1}) \\ \rightarrow \rho_t - \rho_{t-1} &= \kappa\mu_\rho - \kappa\rho_{t-1} \\ \rightarrow Y &= \alpha + \beta X \end{aligned}$$

### 1.9.2. 基础题

**Q-58.** In finance, every risk is also an opportunity. The traders try to forecast changes in correlation and attempt to financially gain from these changes in correlation. What should a trader do, if he expects the correlation to rise in the future?

- I. Enter correlation swaps as a party to pay realized correlation
- II. Buying put options on an index and selling put options on individual components
- III. Paying fixed in a variance swap on an index and receiving fixed on individual components

- A. I and II
- B. II and III
- C. III
- D. All

**Q-59.** Which of the following statements about correlation and copula are correct?

- I. Copula enables the structures of correlation between variables to be calculated separately from their marginal distributions.
  - II. Transformation of variables does not change their correlation structure.
  - III. Correlation can be a useful measure of the relationship between variables drawn from a distribution without a defined variance.
  - IV. Correlation is a good measure of dependence when the measured variables are distributed as multivariate elliptical.
- A. I and IV only
  - B. II, III, and IV only
  - C. I and III only
  - D. II and IV only

**Q-60.** A risk manager uses the past 480 months of correlation data from the Dow Jones Industrial Average (Dow) to estimate the long-run mean correlation of common stocks and the mean reversion rate. Based on historical data, the long-run mean correlation of Dow stocks was 32%, and his regression output estimates the following regression relationship:  $Y = 0.24 - 0.75 X$ . Suppose that in April 2014, the average monthly correlation for all Dow stocks was 36%. What is the expected correlation for May 2014 assuming the mean reversion rate estimated in the regression analysis?

- A. 32%
- B. 33%
- C. 35%
- D. 37%

**Q-61.** A risk manager uses the past 480 months of correlation data from the Dow Jones Industrial Average (Dow) to estimate the long-run mean correlation of common stocks and the mean reversion rate. Based on historical data, the long-run mean correlation of Dow stocks was 34%, and he regression output estimates the following regression relationship:  $Y = 0.262 - 0.77X$ . Suppose that in April 2014, the average monthly correlation for all Dow stocks was 33%. What is the estimated one-period autocorrelation for this time period based on the mean reversion rate estimated in the regression analysis?

- A. 23%
- B. 26%
- C. 30%
- D. 33%

**Q-62.** An investor purchases \$1 million of Canadian bonds and is concerned about the bonds defaulting. The investor wishes to transfer this default risk to a third party, so he enters a fixed credit default swap (CDS) spread agreement with First Bank. Assume that the recovery rate is zero and that there is no accrued interest in the event that Canada defaults. Which of the following statements about the CDS between the investor and First Bank is correct?

- A. A paper loss occurs for the investor if the correlation risk between First Bank and Canada increases because the value of the CDS spread will increase.
- B. The fixed CDS spread is valued based on the default probability of the reference asset and the joint default correlation of First Bank and Canada.
- C. The investor has wrong-way risk if there is negative correlation risk between First Bank and Canada.
- D. Increasing correlation risk decreases the probability that the worst case scenario occurs.

**Q-63.** The dependence structure between the returns of financial assets plays an important role in risk measurement. For liquid markets, which of following statements is incorrect?

- A. Correlation is a valid measure of dependence between random variables for only certain types of return distributions.
- B. Even if the return distributions of two assets have a correlation of zero, the returns of these assets are not necessarily independent.
- C. Copulas make it possible to model marginal distributions and the dependence structure separately.
- D. With short time horizons (3 months or less), correlation estimates are typically very stable.

**Q-64.** A hedge fund that runs a distressed securities strategy is evaluating the solvency conditions of two potential investment targets. Currently firm RST is rated BB and firm WYZ is rated B. The hedge fund is interested in determining the joint default probability of the two firms over the next two years using the Gaussian default time copula under the assumption that a one-year Gaussian default correlation is 0.36. The fund reports that  $X_{BB}$  and  $X_B$  are abscise values of the bivariate normal distribution presented in the table

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below where  $X_{BB} = N^{-1}(Q_{BB}(t_{BB}))$  and  $X_B = N^{-1}(Q_B(t_B))$  with  $t_{BB}$  and  $t_B$  being the time-to-default of BB-rated and B-rated companies respectively; and  $Q_{BB}$  and  $Q_B$  being the cumulative distribution functions of  $t_{BB}$  and  $t_B$ , respectively; and  $N$  denote the standard normal distribution:

| Default Time in Year | Firm RST Default Probability | Firm RST Cumulative Default Probability $Q_{BB}(t)$ | Firm RST Cumulative Standard Normal Percentiles $N^{-1}(Q_{BB}(t))$ | Firm WYZ Default Probability | Firm WYZ Cumulative Default Probability $Q_B(t)$ | Firm WYZ Cumulative Standard Normal Percentiles $N^{-1}(Q_B(t))$ |
|----------------------|------------------------------|-----------------------------------------------------|---------------------------------------------------------------------|------------------------------|--------------------------------------------------|------------------------------------------------------------------|
| 1                    | 5.21%                        | 5.21%                                               | -1.625                                                              | 19.06%                       | 19.06%                                           | -0.876                                                           |
| 2                    | 6.12%                        | 11.33%                                              | -1.209                                                              | 10.63%                       | 29.69%                                           | -0.533                                                           |
| 3                    | 5.50%                        | 16.83%                                              | -0.961                                                              | 8.24%                        | 37.93%                                           | -0.307                                                           |
| 4                    | 4.81%                        | 21.64%                                              | -0.784                                                              | 6.10%                        | 44.03%                                           | -0.150                                                           |
| 5                    | 4.22%                        | 25.86%                                              | -0.648                                                              | 4.03%                        | 48.06%                                           | -0.049                                                           |

Applying the Gaussian copula, which of the following corresponds to the joint probability that firm RST and firm WYZ will both default before the end of year 2?

- A.  $Q(X_{BB} = 0.0612) + Q(X_B = 0.1063) - Q(X_{BB} = 0.0612) \times Q(X_B = 0.1063)$
- B.  $Q(X_{BB} = 0.1133) + Q(X_B = 0.2969) - Q(X_{BB} = 0.1133) \times Q(X_B = 0.2969)$
- C.  $Q(X_{BB} \leq 0.1133 \cap X_B \leq 0.2969)$
- D.  $Q(X_{BB} \leq -1.209 \cap X_B \leq -0.533)$

**Q-65.** A risk analyst at a large investment bank is examining how financial correlation risk affects the bank's portfolios. The bank holds portfolios consisting of different types of assets and enters into various hedging contracts with multiple counterparties. Which of the following statements would the analyst be correct to make?

- A. The buyer of a CDS faces wrong-way risk when there is a positive default correlation between the reference asset and the CDS counterparty.
- B. The risk-adjusted return of a portfolio typically increases when correlations of assets in the portfolio increase.
- C. Dynamic correlation risk in a portfolio of pairs trades is most appropriately estimated using Gaussian copulas.
- D. Correlation risk is highest during periods of relatively benign market movements when correlations are difficult to predict.

- Q-66.** Suppose a correlation swap buyer pays a fixed correlation rate of 0.2 with a notional value of one million U.S. dollar for one year for a portfolio of three assets. The realized pairwise correlations of the daily log returns at maturity for the three assets have correlation matrix below. How much will the buyer of correlation swap receive?

|   | 1   | 2   | 3   | 4   |
|---|-----|-----|-----|-----|
| 1 | 1   | 0.4 | 0.2 | 0.1 |
| 2 | 0.4 | 1   | 0.3 | 0.2 |
| 3 | 0.2 | 0.3 | 1   | 0.2 |
| 4 | 0.1 | 0.2 | 0.2 | 1   |

- A. \$513,954  
 B. \$1,133,333  
 C. \$466,667  
 D. \$233,333
- Q-67.** Which of the following statements is true regarding the correlation level and correlation volatility with respect to the state of economy?
- A. The correlation level is lowest during normal economic period.  
 B. The correlation level increases during recession period.  
 C. The correlation volatility is lowest during normal economic period.  
 D. The correlation volatility is highest during recession period.

## 1.10. Empirical Approaches to Hedging

### 1.10.1. 重要知识点

**1.10.1.1.** Denoting the face amounts of the original position and hedging instrument by  $F_{\text{Position}}$  and  $F_{\text{Instrument}}$ , with DV01s of  $DV01_{\text{Position}}$  and  $DV01_{\text{Instrument}}$ . Our objective is to find the face amount of hedging instrument, which is:

$$F_{\text{Instrument}} = -F_{\text{Position}} \frac{DV01_{\text{Position}}}{DV01_{\text{Instrument}}} \hat{\beta}$$

where  $\hat{\beta}$  is obtained from the regression:

$$\Delta y_t^{\text{Position}} = \alpha + \beta \Delta y_t^{\text{Instrument}} + \varepsilon_t$$

**1.10.1.2.** The hedge of the first equation minimizes the variance of the P&L in over the data set and used to estimate the regression parameters.

**1.10.1.3. Reverse regression:** the reverse regression flips the role of the dependent and independent variables.

**1.10.2. 基础题**

- Q-68.** Assume that a trader is making a relative value trade, selling a U.S. Treasury bond and correspondingly purchasing a U.S. Treasury TIPS. Based on the current spread between the two securities, the trader shorts \$100 million of the nominal bond and purchases \$89.8 million of TIPS. The trader then starts to question the amount of the hedge due to changes in yields on TIPS in relation to nominal bonds. He runs a regression and determines from the output that the nominal yield changes by 1.0274 basis points per basis point change in the real yield. Would the trader adjust the hedge, and if so, by how much?
- A. No
  - B. Yes, by \$2.46 million (purchase additional TIPS).
  - C. Yes, by \$2.5 million (sell a portion of the TIPS).
  - D. Yes, by \$2.11 million (purchase additional TIPS)
- Q-69.** Assume that a trader wishes to set up a hedge such that he sells \$100,000 of a Treasury bond and buys Treasury TIPS as a hedge. Using a historical yield regression framework, assume the DV01 on the T-bond is 0.072, the DV01 on the TIPS is 0.051, and the hedge adjustment factor (regression beta coefficient) is 1.2. What is the face value of the offsetting TIPS position needed to carry out this regression hedge?
- A. \$138,462
  - B. \$169,412
  - C. \$268,499
  - D. \$280,067
- Q-70.** Assumes that bond trader, Jayme Ryan, wishes to make a relative value trade by selling a U.S. Treasury bond (T-bond) and purchasing a U.S. Treasury Inflation Protected Security (TIPS). Ryan decides to short \$100 million of the nominal bond and determines that the DV01 of the TIPS is 0.088 and the DV01 of the T-bond is 0.065. Ryan then runs a least squares regression based on changes in the nominal yield and real yield and finds a yield beta of 1.03 basis points. What is the amount of TIPS that Ryan should purchase in order to hedge the short nominal bond?
- A. \$73.86 million
  - B. \$76.08 million
  - C. \$135.38 million
  - D. \$139.45 million
- Q-71.** A US bank risk manager is conducting a regression hedge to protect a corporate bond position with a face value of \$100m. The regression results indicate that the manager

should sell the US Treasury with a face value of \$93.6m. The colleague of risk manager reviews the hedging procedure and states that "...if you want to hedge a US Treasury with a face value of \$93.6m, then just sell \$100m corporate bond...". Which of the following statements is correct regarding the colleague's statement?

- A. The colleague's statement is correct because the calculated risk weights will be the same under the two hedging transactions.
- B. The colleague's statement is correct because both hedging transactions will minimize the volatility of the hedged position's Profit/Loss.
- C. The colleague's statement is incorrect because the risk of hedging transactions is set by the different face values decided by the hedger.
- D. The colleague's statement is incorrect because these trades are only indifferent if we scale up the reverse regression hedge.

### 1.11. Arbitrage Pricing with Interest Rate (Binomial) Tree

#### 1.11.1. 重要知识点

**1.11.1.1. Arbitrage Pricing of Derivatives:** A security is priced by arbitrage by finding and pricing its replicating portfolio. A portfolio of zero bonds can be used to replicate the option payoff.

**1.11.1.2. Risk-Neutral Pricing:** Risk-neutral pricing is a technique that modifies the assumed interest rate process so that any contingent claim can be priced without having to construct and price its replicating portfolio. The risk-neutral probabilities makes the expected discounted value to equal the market price.

#### 1.11.2. 基础题

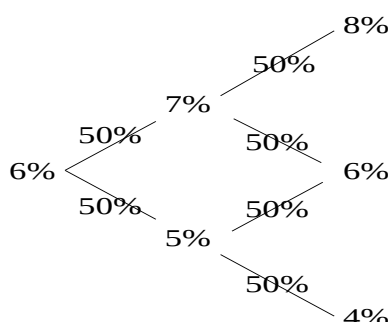
**Q-72.** A European put option has two years to expiration and a strike price of \$101.00. The underlying is a 7% annual coupon bond with three years to maturity. Assume that the risk-neutral probability of an up move is 0.76 in year 1 and 0.60 in year 2. The current interest rate is 3.00%. At the end of year 1, the rate will either be 5.99% or 4.44%. If the rate in year 1 is 5.99%, it will either rise to 8.56% or rise to 6.34% in year 2. If the rate in one year is 4.44%, it will either rise to 6.34% or rise to 4.70%. The value of the put option today is closet to:

- A. \$1.17
- B. \$1.30
- C. \$1.49
- D. \$1.98

**Q-73.** Suppose investors have interest rate expectations as illustrated in the decision tree below

33-80

where the 1-year rate is expected to be 8%, 6%, or 4% in the second year and either 7% or 5% in the first year for a zero-coupon bond.



If investors are risk-neutral, what is the price of a \$1 face value 2-year zero-coupon bond today?

- A. \$0.88113
- B. \$0.88634
- C. \$0.89007
- D. \$0.89032

## 1.12. Expectations, Convexity, Risk Premium, and the Shape of the Term Structure

### 1.12.1. 重要知识点

**1.12.1.1. Expectation:** with investor certainty as to future interest rates, the term structure of forward rates is completely determined by expectations.

**1.12.1.2. Volatility and Convexity:** the interaction of interest rate volatility with the convexity of bond prices drives the forward rates curve down. This arises from a special case of Jensen's inequality:

$$E\left[\frac{1}{1+r}\right] > \frac{1}{1+E[r]}$$

**1.12.1.3. Risk Premium:** the compensation for bearing interest rate risk pushes up the term structure of forward rates.

$$E\left[\frac{dP}{P}\right] = (r_0 + \text{Risk Premium})dt = (r_0 + \lambda D)dt$$

**1.12.1.4. Return Decomposition based on Ito's Lemma:**

$$\frac{dP}{P} = f(T)dt - Ddr + \frac{1}{2}C\sigma^2 dt$$

Return due to the passage of time:  $f(T)dt$ .

Return due to changes in the rate:  $-Ddr$  and  $\frac{1}{2}C\sigma^2 dt$ .

### 1.12.2. 基础题

**Q-74.** An investor expects the current 1-year rate for a zero-coupon bond to remain at 6%, the 1-year rate next year to be 8%, and the 1-year rate in two years to be 10%. What is the 3-year spot rate for zero-coupon bond with face value of \$1, assuming all investor have the same expectations of future 1-year rates for zero-coupon bonds?

- A. 7.888%
- B. 7.988%
- C. 8.000%
- D. 8.088%

**Q-75.** Suppose an investor expects that the 1-year rate will remain at 6% for the first year for a 2-year zero-coupon bond. The investor also projects a 50% probability that the 1-year spot rate will be 8% in one year and a 50% probability that the 1-year spot rate will be 4% in one year. Which of the following inequalities most accurately reflects the convexity effect for this 2-year bond using Jensen's inequality formula?

- A. \$0.89031 > \$0.89000
- B. \$0.89000 > \$0.80000
- C. \$0.94340 > \$0.89031
- D. \$0.94373 > \$0.94340

**Q-76.** Denote the time- $t$  price of a  $T$ -year zero bond by  $P_t(r_t, T)$ , where  $r_t$  is the instantaneous spot rate. By Ito's Lemma,

$$dP = \frac{\partial P}{\partial t} dt + \frac{\partial P}{\partial r} dr + \frac{1}{2} \frac{\partial^2 P}{\partial r^2} \sigma^2 dt$$

We can obtain a simple decomposition of the bond's return. Which of the following statements is correct regarding the decomposition?

- A. The decomposition result states that the return is due to the passage of time, the changes in the rate, and the changes in the spread.
- B. The decomposition result states that the decreases in rate reduce bond return in proportion to duration.
- C. The decomposition result states that the return due to the passage of time is the instantaneous spot rate.
- D. The decomposition result states that lowering rate volatility reduces bond return in proportion to convexity.

**Q-77.** An investor is computing the bond's risk premium. The following information is collected:

|                                      |    |
|--------------------------------------|----|
| Current short-term rate (annualized) | 2% |
|--------------------------------------|----|

|                                        |        |
|----------------------------------------|--------|
| Bond duration                          | 4.2    |
| Risk premium per year of duration risk | 15 bps |

The investor decides to hold the bond for a 6-month investment horizon. Based on the information available, what is the bond's expected return over the specified period?

- A. 1.32%.
- B. 1.63%.
- C. 2.15%.
- D. 2.63%.

### 1.13. One-Factor Term Structure Models

#### 1.13.1. 重要知识点

**1.13.1.1. Model 1:** assumes no drift and that interest rates are normally distributed:

$$dr = \sigma dW_t$$

$$dW_t = z\sqrt{dt} \sim N(0, dt)$$

**1.13.1.2. Model 2:** adds a positive drift term to Model 1 that can be interpreted as a positive risk premium associated with longer time horizons:

$$dr = \lambda dt + \sigma dW_t$$

**1.13.1.3. Ho-Lee Model:** generalizes drift to incorporate time-dependency:

$$dr = \lambda_t dt + \sigma dW_t$$

**1.13.1.4. Vasicek Model:** assumes a mean-reverting process for short-term interest rates:

$$dr = \kappa(\theta - r)dt + \sigma dW_t$$

where:

- $\kappa$  = a parameter that measures the speed of reversion adjustment.
- $\theta$  = long-run value of the short-term rate assuming risk neutrality.
- $r$  = current interest rate level.

$$\text{Half-life: } \tau = \frac{\ln(2)}{\kappa}$$

The interest rate factor takes  $\tau$  years to cover half the distance between its starting value and its goal.

**1.13.1.5. Model 3:** assigns a specific parameterization of time-dependent volatility:

$$dr = \lambda_t dt + \sigma e^{-\alpha t} dW_t$$

where:

- $\sigma$  = volatility at  $t = 0$ , which decreases exponentially to 0 for  $\alpha > 0$

**1.13.1.6. Cox-Ingersoll-Ross (CIR) Model:** mean-reverting model with constant volatility,  $\sigma$ , and basis-point volatility,  $\sigma\sqrt{r}$ , that increases at a decreasing rate:

$$dr = \kappa(\theta - r)dt + \sigma\sqrt{r}dW_t$$

**1.13.1.7. Model 4 (lognormal model):** yield volatility,  $\sigma$ , is constant, but basis-point volatility,  $\sigma r$ , increases with the level of the short-term rate.

$$dr = \alpha r dt + \sigma r dW_t$$

There are two advanced models of importance:

Lognormal with deterministic drift (Salomon Brothers Model):

$$d[\ln(r)] = \alpha_t dt + \sigma dW_t$$

Lognormal with mean reversion (Black-Karasinski Model)

$$d[\ln(r)] = \kappa_t [\ln(\theta_t) - \ln(r)] dt + \sigma_t dW_t$$

### 1.13.2. 基础题

**Q-78.** Model 1 assumes zero drift and is also called a normal model. Model 2 add a term for drift.

Each of the following is true about these two models except for:

- A. A weakness of Model 1 is that the short-term rate can become negative.
- B. Model 1 implies a term structure that is perfectly flat at the current rate for all maturities, including the long-term rates.
- C. Model 2 is more capable of producing an upward-sloping term structure, which is often observed.
- D. Model 2 is an equilibrium model, rather than an arbitrage-free model, because no attempt is made to match the term structure closely.

**Q-79.** A risk manager is pricing a 10-year Treasuries using a successfully tested pricing model. Current interest rate volatility is high and the risk manager is concerned about the effect this may have on short-term rates when pricing the option. Which of the following actions would best address the potential for negative short-term interest rates to arise in the model?

- A. The risk manager uses a normal distribution of interest rates.
- B. When short-term rates are negative, the risk manager adjusts the risk-neutral probabilities.
- C. When short-term rates are negative, the risk manager increases the volatility.
- D. When short-term rates are negative, the risk manager sets the rate to zero.

**Q-80.** John Jones, FRM, is discussing the appropriate usage of mean-reverting models relative to no-drift models, models that incorporate drift, and Ho-Lee models. Jones makes the following statements:

- Statement 1: Both Model 1 (no drift) and the Vasicek model assume parallel shifts from changes in the short-term rate.
- Statement 2: The Vasicek model assumes decreasing volatility of future short-term

rates while Model 1 assumes constant volatility of future short-term rates.

- Statement 3: The constant drift model (Model 2) is a more flexible model than the Ho-Lee model.

How many of his statements are correct?

- A. 0
- B. 1
- C. 2
- D. 3

**Q-81.** Using Model 1, assume the current short-term interest rate is 5%, annual volatility is 80bps, and  $dW$ , a normally distribution random variable with mean 0 and standard deviation  $\sqrt{dt}$ , has an expected value of zero. After one month, the realization of  $dW$  is -0.5. What is the change in the spot rate and the new spot rate?

|    | Change in Spot | New Spot Rate |
|----|----------------|---------------|
| A. | 0.40%          | 5.40%         |
| B. | -0.40%         | 4.60%         |
| C. | 0.80%          | 5.80%         |
| D. | -0.80%         | 4.20%         |

**Q-82.** An analyst is modeling spot rate changes using short rate term structure models. The current short-term interest rate is 5% with a volatility of 80 bps. After one month passes the realization of  $dW$ , a normally distributed random variable with mean 0 and standard deviation  $\sqrt{dt}$ , is -0.5. Assume a constant interest rate drift,  $\lambda$ , of 0.36%. What should the analyst compute as the new spot rate?

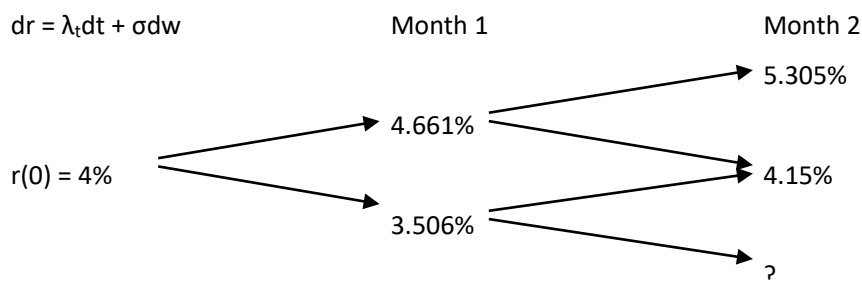
- A. 5.37%
- B. 4.63%
- C. 5.76%
- D. 4.24%

**Q-83.** The current short-term rate,  $r(0)$  is 4%. Under a Ho-Lee Model with time-dependent drift, the time step is monthly and the annualized drifts are as follows: +100 basis points in the first month and +80 basis points in the second month. The annual basis point volatility is 200bps.

#### Ho-Lee Assumptions

|                                            |         |
|--------------------------------------------|---------|
| Month ( $dt$ )                             | 0.0833  |
| Annualized drift, first month, $\lambda_1$ | 100 bps |

|                                             |        |
|---------------------------------------------|--------|
| Annualized drift, second month, $\lambda_2$ | 80 bps |
| Volatility, annual                          | 2%     |
| Ho-Lee Model: Time-dependent drift          |        |



What is the value of the missing node  $[2, 0]$  in this Ho-Lee interest rate tree?

- A. 2.447%
- B. 2.677%
- C. 2.995%
- D. 3.256%

**Q-84.** Which of the following statements best characterizes the differences between the Ho-Lee model with drift and the lognormal model with drift?

- A. In the Ho-Lee model and the lognormal model the drift terms are multiplicative.
- B. In the Ho-Lee model and the lognormal model the drift terms are additive
- C. In the Ho-Lee model the drift terms are multiplicative, but in the lognormal model the drift terms are additive
- D. In the Ho-Lee model the drift terms are additive, but in the lognormal model the drift terms are multiplicative.

**Q-85.** A CRO of a hedge fund is asking the risk team to develop a term-structure model that is appropriate for fitting interest rates for use in the fund's options pricing practice. The risk team is evaluating among several interest rate models with time-dependent drift and time-dependent volatility functions. Which of the following is a correct description of the specified model?

- A. In the Ho-Lee model, the drift of the interest rate process is presumed to be constant.
- B. In the Ho-Lee model, when the short-term rate is above its long-run equilibrium value, the drift is presumed to be negative.
- C. In the Cox-Ingersoll-Ross model, the basis-point volatility of the short-term rate is presumed to be proportional to the square root of the rate, and short-term rates cannot be negative.
- D. In the Cox-Ingersoll-Ross model, the volatility of the short-term rate is presumed to

decline exponentially to a constant level.

- Q-86.** The trading desk at Big Bank is pricing an off-market swap. The quantitative analysis team has identified the interest rate drift in periods 1 and 2 to be 25 basis points and -10 basis points, respectively. These values were calibrated from liquid swap prices. The current short-term interest rate is 5.4% with a long-run mean reverting level of 15.1%. Additionally, the long-run true interest rate is 12.6%. The time steps is 1; i.e.,  $dt=1$ . Using the HO-LEE Model, what is the expected short-term interest rate in two periods?
- A. 5.25%
  - B. 5.4%
  - C. 5.55%
  - D. 5.75%
- Q-87.** An analyst is looking at various models used to incorporate drift into term structure models. The Ho-Lee Model:
- A. Incorporates no-risk premium to the interest rate model allowing rates to vary according to their volatility.
  - B. Incorporates drift as a premium to interest rates that remains constant over time.
  - C. Allows for a risk premium to be applied to interest rates that changes over time.
  - D. Incorporates drift into the model following the assumption that rates revert to the long-run equilibrium value.
- Q-88.** A risk manager is constructing a term structure model and intends to use the Cox-Ingersoll-Ross Model. Which of the following describes this model?
- A. The model presumes that the volatility of the short rate will increase at a predetermined rate.
  - B. The model presumes that the volatility of the short rate will decline exponentially to a constant level.
  - C. The model presumes that the basis-point volatility of the short rate will be proportional to the rate.
  - D. The model presumes that the basis-point volatility of the short rate will be proportional to the square root of the rate.
- Q-89.** A hedge fund risk manager is looking at various models that are flexible enough to incorporate mean reversion and risk premium into term structure modeling. Which of the following is correct about the Vasicek model?
- A. It incorporates mean reversion feature and its drift is always zero.

- B. It incorporates mean reversion feature and models the risk premium as a constant or changing drift.
- C. It cannot incorporate risk premium and the term structure of interest rate volatility in the model is upward-sloping.
- D. It cannot capture the mean reversion feature but can be used to model the time-varying risk premium.

**Q-90.** A financial analyst is pricing a 3-year call option on a 3-year Treasury note using a pricing model that has been successfully validated. Interest rate volatility is currently high, and the analyst is concerned that this may cause the model to forecast negative short-term interest rates, which would in turn cause the option price to be misvalued. Which of the following actions would be best for the analyst to take to address negative short-term interest rates when they arise in the model?

- A. Adjusting the risk-neutral probabilities
- B. Increasing the volatility
- C. Setting the interest rate to zero
- D. Setting the mean-reverting parameter to 1

**Q-91.** A senior risk manager at a hedge fund company is evaluating methods to improve the fund's ability to model interest rate term structures. The manager would like to adopt a model that is flexible enough to incorporate both mean reversion and a risk premium and considers the Vasicek model for this purpose. Which of the following statements is correct about the Vasicek model?

- A. The model incorporates mean reversion and a drift of zero.
- B. The model incorporates mean reversion and models the risk premium as a component of the drift.
- C. The model incorporates a drift of zero, but it cannot incorporate a risk premium.
- D. The model can be used to model the time-varying risk premium, but it cannot incorporate mean reversion.

**Q-92.** An analyst uses interest-rate trees to forecast short-term interest rates. The analyst applies the following model for estimating monthly changes in a short-term interest rate tree:

$$dr = \lambda(t) \cdot dt + \sigma(t) \cdot dw$$

In this process,  $\lambda(t)$  represents the drift in month  $t$ ,  $\sigma(t)$  represents the volatility in month  $t$ ,  $dt$  is the time interval measured in years, and  $dw$  is a normally distributed random variable with a mean of zero and a standard deviation of the square root of  $dt$ . The analyst uses the following information to make the calculations:

- Current level of short-term interest rate: 3.1%
- Drift in month 1 ( $\lambda(1)$ ): 0.0024
- Drift in month 2 ( $\lambda(2)$ ): 0.0036
- Annualized volatility of the interest rate in month 1 ( $\sigma(1)$ ): 0.0060
- Annualized volatility of the interest rate in month 2 ( $\sigma(2)$ ): 0.0080
- Probability of an upward or downward movement in interest rates: 0.5

What is the volatility component of the change in interest rate from the upper node of month 1 to the upper node of month 2?

- A. 23 bps
- B. 26 bps
- C. 40 bps
- D. 45 bps

**Q-93.** An analyst is studying term structure models that incorporate measures of volatility into the interest rate process. The analyst focuses on the Cox-Ingersoll-Ross (CIR) model and its treatment of volatility of the short-term interest rate. Which of the following statements is correct regarding the yield volatility and basis-point volatility in the CIR model?

- A. Periods of extremely low short-term interest rates are accompanied by high basis-point volatility, which increases the possibility of negative interest rates in the CIR model.
- B. In the CIR model, basis-point volatility is specified as a decreasing function of the mean reversion factor.
- C. In the CIR model, yield volatility is specified as being constant while basis-point volatility is allowed to vary.
- D. Basis-point volatility and yield volatility are used interchangeably to measure the same volatility in the CIR model.

**Q-94.** A quantitative analyst on the fixed-income desk of an investment bank is applying the Vasicek model to estimate future short-term interest rates. The model is given below:

$$dr = k * (\Theta - r) * dt + \sigma * dw$$

where  $dr$  is the change in the short-term interest rate,  $\Theta$  is the estimated long-run value of the short-term interest rate,  $k$  is the mean reversion rate,  $r$  is the current level of the short-term interest rate,  $\sigma$  is the annual basis-point volatility of the short-term interest rate,  $dt$  is the time interval measured in years, and  $dw$  is a normally distributed random variable with mean zero and standard deviation equal to the square root of  $dt$ .

The analyst gathers the following information:

- Current short-term interest rate ( $r$ ): 3.35%
- Long-run value of short-term interest rate ( $\Theta$ ): 4.55%

- Mean reversion rate ( $k$ ): 0.06
- Annual basis-point volatility ( $\sigma$ ): 120 bps

The analyst then creates an interest rate tree, determines the expected short-term interest rate after 8 years, and calculates how long it will take the short-term interest rate to revert halfway to the long-run value. Which of the following statements would be correct for the analyst to make?

- The expected short-term interest rate is 3.81% and the half-life is 11.6 years.
- The expected short-term interest rate is 3.81% and the half-life is 16.7 years.
- The expected short-term interest rate is 4.09% and the half-life is 11.6 years.
- The expected short-term interest rate is 4.09% and the half-life is 16.7 years.

## 1.14. The Gauss+ Model

### 1.14.1. 重要知识点

#### 1.14.1.1. Model Specification:

$$dr_t = \alpha_r(m_t - r_t)dt$$

$$dm_t = \alpha_m(l_t - m_t)dt + \sigma_m(\rho dW_t^1 + \sqrt{1 - \rho^2}dW_t^2)$$

$$dl_t = \alpha_l(\mu - l_t)dt + \sigma_l dW_t^1$$

$r_t$ : short-term rate.

$m_t$ : medium-term factor, which reflects business cycles and monetary policy.

$l_t$ : long-term factor, which reflects long-term expectations of inflation and real interest rate, which ultimately depends on long-term trends in demographics, production technology, etc.

$\mu$ : a constant that  $l_t$  mean reverts to.

$\{\alpha_r, \alpha_m, \alpha_l\}$ : mean reversion parameters of factors, where  $\alpha_l < \alpha_m < \alpha_r$ .

$\{\sigma_m, \sigma_l\}$ : volatility parameters for  $m_t$  and  $l_t$ .

$\{dW_t^1, dW_t^2\}$ : two independent Brownian motions.

$\rho$ : correlation among  $dm_t$  and  $dl_t$ .

#### 1.14.1.2. Gauss+ Model vs. Vasicek Model:

| Vasicek Model                                                 | Gauss+ Model                                               |
|---------------------------------------------------------------|------------------------------------------------------------|
| A one-factor model.                                           | A three-factor model (with two sources of risk).           |
| Assumes a normal distribution for the factor.                 | Assumes normal distributions for the factors.              |
| Cannot capture the hump-shaped term structures of volatility. | Can capture the hump-shaped term structures of volatility. |
| A simple but less flexible model.                             | A model that balances the tractability                     |

|  |                  |
|--|------------------|
|  | and flexibility. |
|--|------------------|

**1.14.2. 基础题**

- Q-95.** Which of the following statements correctly describes the dynamics and implications of the cascade form of the Gauss+ model?
- A. Under the Gauss+ model, the short-term rate mean reverts to a medium-term factor, the medium-term factor mean reverts to a long-term factor, and the long-term factor is modelled with time-dependent drift and time-dependent volatility.
  - B. Under the Gauss+ model, the evolution of short-term rate reflects the fact that the Fed keeps the short-term policy rate pegged at a target, and this helps generate a “hump-shaped” term structure of volatility that fits the empirical data quite well.
  - C. Under the Gauss+ model, the evolution of short-term rate and medium-term factor is modelled with a pre-specified correlation parameter, while the long-term factor is assumed to be independent of the medium-term factor.
  - D. Under the Gauss+ model, the speed of mean reversion from short-term to medium-term is assumed to be equal to that from medium-term to long-term, while the empirical evidence suggests that the speed of mean reversion among factors is different.
- Q-96.** There is an incredible improvement from the one-factor model to multifactor model in studying the term structure dynamics. Which of the following statements is correct regarding the well-known one-factor Vasicek model and the multifactor Gauss+ model?
- A. Although the Vasicek model can explain a large fraction of term structure dynamics with merely one factor, it only has some limited uses for practitioners because the model structure is complex.
  - B. The Vasicek model can be used for computing the bond sensitivities to changes in interest rates. This is achieved through shifting the short-rate, re-computing the bond prices and calculating measures such as DV01 and durations.
  - C. The Vasicek model might be used to conduct relative-value trading, while the Gauss+ model is inappropriate for this purpose. However, both models can be thought of as a tool for hedging.
  - D. Although the Gauss+ model captures the empirical complexity of the term structure dynamics, its concepts and assumptions are not intuitive and the model tractability is severely harmed.
- Q-97.** A practitioner collects the recent fixed-income market data and obtain the following estimation results for the Gauss+ model:

| Parameters | Estimation Results |
|------------|--------------------|
| $\alpha_r$ | 1.0547             |
| $\alpha_m$ | 0.6358             |
| $\alpha_l$ | 0.0165             |
| $\sigma_m$ | 109.2 bps          |
| $\sigma_l$ | 96.4 bps           |
| $\rho$     | 0.212              |
| $\mu$      | 10.56%             |
| $r_t$      | 2.25%              |
| $m_t$      | 4.90%              |
| $l_t$      | 8.32%              |

Based on the information provided, what is the expected short-term rate in 3-month?

- A. 1.55%.
- B. 2.95%.
- C. 5.04%.
- D. Further information is required.

**Q-98.** A practitioner collects the recent fixed-income market data and obtain the following estimation results for the Gauss+ model:

| Parameters | Estimation Results |
|------------|--------------------|
| $\alpha_r$ | 1.0547             |
| $\alpha_m$ | 0.6358             |
| $\alpha_l$ | 0.0165             |
| $\sigma_m$ | 109.2 bps          |
| $\sigma_l$ | 96.4 bps           |
| $\rho$     | 0.212              |
| $\mu$      | 10.56%             |
| $r_t$      | 2.25%              |
| $m_t$      | 4.90%              |
| $l_t$      | 8.32%              |

Based on the information provided, what is the half-life for long-term factor?

- A. 0.66 years.
- B. 1.09 years.
- C. 42.01 years.
- D. Further information is required.

- Q-99.** Suppose that under the Gauss+ model, the long-term factor is currently at 3.0% with a volatility parameter of 120 bps. The speed of mean reversion for the factor is estimated to be 0.04, and the realized  $dW_t$  in the long-term factor process is 0.24. Further, you have estimated that the change in long-term factor over a 6-month period is 0.40%. What is the value of the constant that the long-term factor mean reverts to?
- A. 3.00%.  
B. 5.80%.  
C. 8.60%.  
D. Further information is required.
- Q-100.** The practical estimation method for Gauss+ model's parameters proceeds in stages, with each stage estimating a subset of the model's parameters. Which of the following statements is correct regarding the estimation method?
- A. In order to estimate the mean reversion parameters, one should first regress the changes in the zero yields of medium- long-term bonds on the changes in the zero yields of various terms.  
B. In order to estimate the value to which the short-term rate reverts over the very long-term (i.e., the constant  $\mu$ ), we should ensure that the term structure of volatilities in the model matches the term structure of volatilities in the data as closely as possible.  
C. Once estimated, the short-term rate, medium-term factor and long-term factor are set equal to the fed funds target rate, the one-year rate that is two-year forward and the one-year rate that is ten-year forward, respectively.  
D. This practical estimation method is significantly easier to implement than the maximum likelihood methods that are standard in the term structure literature.

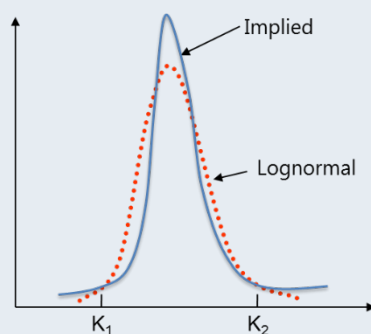
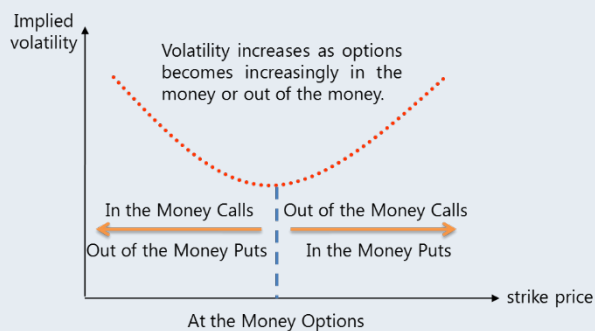
### 1.15. Volatility Smile

#### 1.15.1. 重要知识点

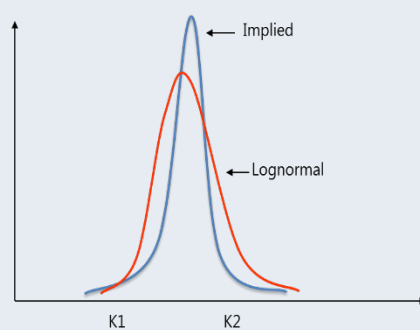
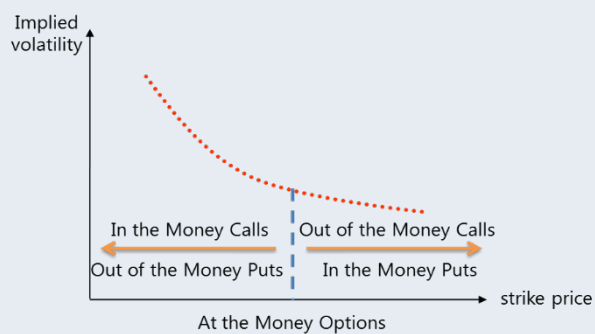
**1.15.1.1.** Put-call parity indicates that the deviation between market prices and Black-Scholes-Merton prices will be equivalent for calls and puts. Hence, implied volatility will be the same for calls and puts.

$$p_{BS} - p_{mkt} = c_{BS} - c_{mkt}$$

#### 1.15.1.2. Volatility Smile for Foreign Currency Options



### 1.15.1.3. Volatility Smiles (skew) for Equity Options



### 1.15.2. 基础题

**Q-101.** The Chief Risk Officer of Martingale Investments Group is planning a change in methodology for some of the risk management models used to estimate risk measures. His aim is to move from models that use the normal distribution of returns to models that

use the distribution of returns implied by market prices. Martingale Group has a large long position in the German equity stock index DAX which has a volatility smile that slopes downward to the right. How will the change in methodology affect the estimate of expected shortfall (ES)?

- A. ES with the updated models will be larger than the old estimate.
- B. ES with the updated models will be smaller than the old estimate.
- C. ES will remain unchanged.
- D. Insufficient information to determine.

**Q-102.** With all other things being equal, a risk monitoring system that assumes constant volatility for equity returns will understate the implied volatility for which of the following positions by the largest amount:

- A. Short position in an at-the-money call
- B. Long position in an at-the-money call
- C. Short position in a deep out-the-money call
- D. Long position in a deep in-the-money call

**Q-103.** Which of the following regarding equity option volatility is true?

- A. There is higher implied price volatility for away-from-the-money equity options.
- B. "Crashophobia" suggests actual equity volatility increases when stock prices decline.
- C. Compared to the lognormal distribution, traders believe the probability of large down movements in price is similar to large up movements.
- D. Increasing leverage at lower equity prices suggests increasing volatility.

**Q-104.** You are asked to mark to market a book of plain vanilla stock options. The trader is short deep out-of-the-money options and long at-the-money options. There is a pronounced smile for these options. The trader's bonus increases as the value of his book increases. Which approach should you use to mark the book?

- A. Use the implied volatility of at-the-money options because the estimation of the volatility is more reliable.
- B. Use the average of the implied volatilities for the traded options for which you have data because all options should have the same implied volatility with Black-Scholes and you don't know which one is the right one.
- C. For each option, use the implied volatility of the most similar option traded on the market.
- D. Use the historical volatility because doing so corrects for the pricing mistakes in the option market.

- Q-105.** The market price of a European call is \$3.00 and its Black-Scholes price is \$3.50. The Black-Scholes price of a European put option with the same strike price and time to maturity is \$2.00. What should the market price of this option be?
- A. \$1.50
  - B. \$2.00
  - C. \$1.00
  - D. \$0.50
- Q-106.** An empirical distribution of equity price derived from the price of options of such stock based on BSM that exhibits a fatter right tail than that of a lognormal distribution would indicate:
- A. Equal implied volatilities across low and high strike prices.
  - B. Greater implied volatilities for low strike prices.
  - C. Greater implied volatilities for high strike prices.
  - D. Higher implied volatilities for mid-range strike prices.
- Q-107.** Compared to at-the-money currency options, out-of-the-money currency options exhibit which of the following volatility traits?
- A. Lower implied volatility
  - B. A frown
  - C. A smirk
  - D. Higher implied volatility
- Q-108.** Equity options tend to exhibit a volatility “smirk” where low strike price options have a higher implied volatility than high strike price options. An explanation that has not been used for the smirk pattern is:
- A. heteroscedasticity in the underlying.
  - B. a higher proportion of debt in the capital structure as equity prices fall.
  - C. less firm leverage as equity prices rise.
  - D. the threat of another market crash.
- Q-109.** Which of the following statement is incorrect regarding volatility smiles?
- A. Currency options exhibit volatility smiles because the at-the-money option have higher implied volatility away-from-the-money options.
  - B. Volatility frowns result when jumps occur in asset prices
  - C. Equity options exhibit a volatility smirk because low strike price options have grater

implied volatility.

- D. Relative to currency traders, it appears that equity traders' expectations of extreme price movements are more asymmetric.

**Q-110.** A risk manager is examining a firm's equity index option price assumptions. The observed volatility skew for a particular equity index slopes downward to the right. Compared to the lognormal distribution, the distribution of option prices on this index implied by the Black-Scholes-Merton (BSM) model would have:

- A. A fat left tail and a thin right tail.
- B. A fat left tail and a fat right tail.
- C. A thin left tail and a fat right tail.
- D. A thin left tail and a thin right tail.

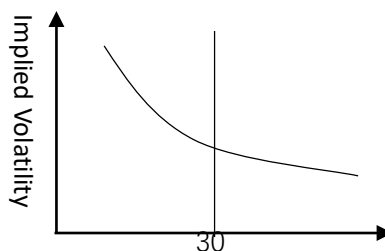
**Q-111.** A committee of risk management practitioner discusses the difference between pricing deep out-of-the-money call options on FBX stock and pricing deep out-of-the-money call options on the EUR/JPY foreign exchange rate using the Black-Scholes-Merton (BSM) model. The practitioners price these options based on two distinct probability distributions of underlying asset prices at the option expiration date:

- A lognormal probability distribution
- An implied risk-neutral probability distribution obtained from the volatility smile for options of the same maturity

Using the lognormal instead of the implied risk-neutral probability distribution will tend to:

- A. Price the option on FBX relatively high and price the option on EUR/JPY relatively low.
- B. Price the option on FBX relatively low and price the option on EUR/JPY relatively low.
- C. Price the option on FBX relatively low and price the option on EUR/JPY relatively high.
- D. Price the option on FBX relatively high and price the option on EUR/JPY relatively high.

**Q-112.** A risk manager is in the process of valuing several European option positions on a non-dividend-paying stock XYZ that is currently priced at GBP 30. The implied volatility skew, estimated using the Black-Scholes-Merton model and the current prices of actively traded European-style options on stock XYZ at various strike prices, is



Assuming that the implied volatility at GBP 30 is used to conduct the valuation, which of the following long positions will be undervalued?

- A. An out-of-the-money call
- B. An in-the-money call
- C. An at-the-money put
- D. An in-the-money put

**Q-113.** A risk analyst at an investment bank is examining the assumptions the bank uses in its foreign exchange (FX) option pricing model. Currently, the implied volatility in a particular FX pair is relatively low if an option is at-the-money and becomes progressively higher as the option moves either more in-the-money or more out-of- the-money. How does the distribution of option prices on this FX pair implied by the Black-Scholes-Merton model compare to a lognormal distribution with the same mean and standard deviation?

- A. The implied distribution has a heavier left tail and a less heavy right tail.
- B. The implied distribution has a heavier left tail and a heavier right tail.
- C. The implied distribution has a less heavy left tail and a heavier right tail.
- D. The implied distribution has a less heavy left tail and a less heavy right tail.

**Q-114.** A risk manager of company ABC is discussing the difference between pricing deep out-of-the-money call options on ABC stock and pricing deep out-of- the-money call options on the USD/GBP foreign exchange (FX) rate using the Black-Scholes-Merton model. The manager considers pricing each of these two options based on two distinct probability distributions of underlying asset prices at the option expiration date: a lognormal probability distribution, and an implied risk- neutral probability distribution obtained from the volatility smile for each aforementioned option of the same maturity and the same moneyness. If the implied risk-neutral probability distribution is used instead of the lognormal distribution, which of the following is correct?

- A. The price of the option on ABC stock would be relatively high and the price of the option on USD/GBP FX rate would be relatively low compared to those computed from the lognormal counterparts.

- B. The price of the option on ABC stock would be relatively low and the price of the option on USD/GBP FX rate would be relatively high compared to those computed from the lognormal counterparts.
- C. The price of the option on ABC stock would be relatively low and the price of the option on USD/GBP FX rate would be relatively low compared to those computed from the lognormal counterparts.
- D. The price of the option on ABC stock would be relatively high and the price of the option on USD/GBP FX rate would be relatively high compared to those computed from the lognormal counterparts.

**Q-115.** A market-maker on the foreign exchange (FX) desk at an investment bank has been asked to provide a quote for an FX call option that expires in 7 months. The option has a strike price ( $K$ ) to spot price ( $S_0$ ) ratio of 1.075. The market-maker references the following implied volatility surface when creating the quote:

| Time to<br>expiration | Strike price to spot price ratio ( $K/S_0$ ) |      |      |      |      |
|-----------------------|----------------------------------------------|------|------|------|------|
|                       | 0.90                                         | 0.95 | 1.00 | 1.05 | 1.10 |
| 1 month               | 9.25                                         | 8.55 | 8.05 | 8.70 | 9.45 |
| 3 months              | 9.10                                         | 8.70 | 8.30 | 8.75 | 9.15 |
| 6 months              | 9.45                                         | 9.05 | 8.70 | 9.10 | 9.45 |
| 1 year                | 9.65                                         | 9.50 | 9.35 | 9.55 | 9.75 |

What implied volatility should the market-maker use to create the quote?

- A. 9.18%
  - B. 9.28%
  - C. 9.34%
  - D. 9.65%
- Q-116.** An option pricing analyst has been asked to write a report examining the relationship between option prices and implied volatility curves. The analyst notes that the implied volatility curves of different underlying assets often have different shapes and explains the reasons why this occurs. Which of the following statements can correctly be included in the report?
- A. The implied volatility smile commonly seen in equity options is due to the higher probability of a greater than three standard deviation price change than would be expected if prices are lognormally distributed.
  - B. The implied volatility smile commonly seen in foreign exchange rate options is due to the higher probability of a price change of between one and two standard deviations from the mean than would be expected if prices are lognormally distributed.

- C. Demand for option protection against steep drops in equity prices leads to higher prices in out-of-the-money puts relative to out-of-the-money calls, which creates a downward-sloping implied volatility skew in these options.
- D. Demand for option protection against the impact of unexpected central bank announcements on foreign exchange rates leads to higher prices, and higher implied volatilities, for at-the-money options relative to out-of-the-money options.

### 1.16. Key Point: Fundamental Review of the Trading Book

#### 1.16.1. 重要知识点

##### 1.16.1.1. Expected Shortfall

The FRTB is proposing a change to the measure used for determining market risk capital. Instead of VaR with a 99% confidence level, expected shortfall (ES) with a 97.5% confidence level is proposed.

#### 1.16.2. 基础题

- Q-117.** Which of the following statements regarding the differences between Basel I, Basel II.5, and the Fundamental Review of the Trading Book (FRTB) for market risk capital calculations is incorrect?
- A. Both Basel I and Basel II.5 require calculation of VaR with a 99% confidence interval.
  - B. FRTB requires the calculation of expected shortfall with a 97.5% confidence interval.
  - C. FRTB requires adding a stressed VaR measure to complement the expected shortfall calculation.
  - D. The 10-day time horizon for market risk capital proposed under Basel I incorporates a recent period of time, which typically ranges from one to four years.
- Q-118.** Which of the following risks is specifically recognized by the incremental risk charge (IRC)?
- A. Expected shortfall risk, because it is important to understand the amount of loss potential in the tail.
  - B. Jump-to-default risk, as measured by 99.9% VaR, because a default could cause a significant loss for the bank.
  - C. Equity price risk, because a change in market prices could materially impact mark-to-market accounting for risk.
  - D. Interest rate risk, as measured by 97.5% expected shortfall, because an increase in interest rates could cause a significant loss for the bank.
- Q-119.** A regulatory analyst at a multinational bank is examining regulatory requirements the bank

must comply with under the Basel Committee's FRTB guidelines. The analyst explores how the FRTB guidelines evolved from the Basel I and Basel II.5 frameworks as well as the instructions for applying the guidelines. Which of the following is correct regarding the FRTB?

- A. While Basel I and Basel II.5 allowed market risk to be calculated at the trading desk level, FRTB requires that market risk be calculated on a firm-wide basis.
- B. While Basel I and Basel II.5 emphasized the use of a standardized approach to calculating market risk, FRTB encourages each bank to develop and rely on an internal models approach.
- C. FRTB standardizes the liquidity horizon used for all risk factors in the market risk capital calculation as 10 days, rather than the different horizons used in Basel I and Basel II.5.
- D. FRTB requires that the stressed ES measure be used in determining market risk capital, rather than the VaR and stressed VaR measures that were used in Basel I and Basel II.5, respectively.

**Q-120.** A quantitative risk analyst at a bank has been asked to incorporate market liquidity into the bank's VaR model. The analyst examines how changes in the level of market liquidity impact the liquidity horizon and the bank's credit risk exposure. Which of the following statements describes the most likely impact of changes in market liquidity?

- A. If market liquidity decreases, the liquidity horizon will shorten, and the credit risk exposure will increase.
- B. If market liquidity decreases, the liquidity horizon will lengthen, and the credit risk exposure will increase.
- C. If market liquidity increases, the liquidity horizon will shorten, and the credit risk exposure will increase.
- D. If market liquidity increases, the liquidity horizon will lengthen, and the credit risk exposure will decrease.

## Solution

### Q-1. Solution: D

### Q-2. Solution: B

B is correct. Non-parametric approaches can accommodate fat tails, skewness, and any other non-normal features that can cause problems for parametric approaches. However, if the data period that is used in estimation includes few losses or losses with low magnitude, non-parametric methods will often produce inaccurate risk measures. Also, non-parametric approaches produce VaR and ES that are too low if the data period has unusually low volatility, and would produce VaR and ES that are too high if the data period has unusually high volatility. Hence parametric methods would be more appropriate in those situations. Therefore, A, C and D are incorrect.

### Q-3. Solution: C

We need to map the portfolio to a position in the underlying stock TUV. A deep in-the-money call has a delta of approximately 1, a deep out-of-the-money call has a delta of approximately zero and forwards have a delta of 1. The net portfolio has a delta of about  $1 \times 5,000 + 0 \times 20,000 + 1 \times 10,000 = 15,000$  and is approximately gamma neutral.

Let:

$\alpha = 2.326$  (99% confidence level)

$S =$  price per share of stock TUV = USD 52

$D_p =$  delta of the position = 15,000

$\sigma =$  volatility of TUV = 0.12

Therefore, the 1-day VaR estimate at 99% confidence level is computed as follows:

$$2.326 \times 52 \times 15,000 \times 0.12 \times \sqrt{1/252} = 13,714.67$$

### Q-4. Solution: A

$$VaR_{Mic} = 1.65 \times 2\% \times 120 \times 1000 = 3960$$

$$VaR_{AT\&T} = 1.65 \times 1\% \times 30 \times 20000 = 9900$$

$$VaR_{P(5\text{-day}, 95\%)} = \sqrt{3960^2 + 9900^2 + 2 \times 0.3 \times 3960 \times 9900} \times \sqrt{5} = 26193$$

### Q-5. Solution: C

$$\text{Normal VaR} = |0.1 - 1.645 \times 0.4| = 0.558$$

$$\text{Lognormal VaR} = 1 - e^{0.1 - 1.645 \times 0.4} = 0.4276$$

Hence, lognormal VaR is smaller than Normal VaR by 13.04% per year. With a portfolio of GBP 1,000,000, this translates to GBP 130,400.

**Q-6. Solution: A**

VaR does not consider losses beyond the VaR threshold level.

**Q-7. Solution: C**

An estimate of the expected shortfall (ES) can be obtained by taking the average of the VaRs for the various confidence levels that are greater than 97.5%. Therefore,

$$ES = (378,412,500 + 392,452,500 + 410,880,000 + 439,252,500) / 4 = \text{JPY } 405,249,375.$$

**Q-8. Solution: B**

Expected shortfall is always greater than or equal to VaR for a given confidence level, since ES accounts for the severity of expected losses beyond a particular confidence level, while VaR measures the minimum expected loss at that confidence level. Therefore, ES would lead to a higher level of required economic capital than VaR for the same confidence level. In practice, however, regulators often correct for the difference between ES and VaR by lowering the required confidence level for banks using ES compared to those using VaR.

**Q-9. Solution: C**

The expected shortfall is an estimate of the tail loss by averaging the VaRs for increasing confidence levels. The tail is divided into  $n$  slices and the corresponding  $n - 1$  VaRs are calculated. In this case,  $n = 5$ , so only the VaRs for the 96%, 97%, 98%, and 99% levels are averaged to calculate expected shortfall.  $(1.7507 + 1.8808 + 2.0537 + 2.3263) / (5 - 1) = 2.003$ .

**Q-10. Solution: C**

Heavy tails: the steeper slope - i.e., greater than 1.0 - at the tails indicates the tails are heavier than the reference distribution.

**Q-11. Solution: D**

This Q-Q plot has steeper slopes at the tails of the plot, which indicate fat tails in the distribution. A normal distribution would result in a linear QQ plot. A distribution with thin tails would produce a QQ plot with less steep slopes at the tails of the plot than a linear relationship, while this one is steeper at the tails. It is not a negatively skewed distribution, as the Q-Q plot is symmetric.

**Q-12. Solution: A**

A is correct. If the empirical distribution has heavier tails than the reference distribution, the QQ plot will have steeper slopes at its tails. Conversely, if the empirical distribution has lighter tails than the reference distribution, the QQ plot will have flatter slopes at its tails.

B is incorrect. Outliers are easily identified on QQ plots due to the nature of the plotting, and this

is one of the ways QQ plots are useful.

C is incorrect. Reference distributions that produce non-linear QQ plots can be dismissed from consideration as a good representative of the empirical data since data drawn from the reference distribution should produce a linear QQ plot .

D is incorrect. The primary purpose of a QQ plot is to determine the proper reference distribution. One can see if there is skewness and kurtosis although it cannot be measured with a QQ plot.

**Q-13. Solution: D**

D is correct. The halving error and the standard error of a quantile estimator will both decrease as the number of slices,  $n$ , or quantiles, used in the process of estimating the risk measure increases .

A is incorrect. Coherent risk measures can be created using sophisticated weighting functions and do not need to be equally weighted.

B is incorrect. The building blocks of quantile estimation are essentially the same needed to estimate coherent risk measures.

C is incorrect. QQ plots are a useful diagnostic tool when considering what kind of distribution might fit the profit and loss data. QQ plots are not used to analyze or evaluate quantile estimators.

**Q-14. Solution: B**

Bootstrapping from historical simulation involves repeated sampling with replacement. The 5% VaR is recorded from each sample draw. The average of the VaRs from all the draws is the VaR estimate. The bootstrapping procedure does not involve filtering the data or weighting observations. Note that the VaR from the original data set is not used in the analysis.

**Q-15. Solution: D**

The historical return is adjusted based on the ratio of current daily volatility to historically observed daily volatility 30 days ago. The volatility-adjusted return is calculated as follows:

$$r_{t,i}^* = \frac{\sigma_{T,i}}{\sigma_{t,i}} * r_{t,i} = \frac{0.014}{0.01} * 0.015 = 0.021 = 2.1\%$$

Once the volatility-adjusted return is computed, VAR, ES, and any other coherent risk measure can be calculated in the usual way after substituting historical returns with volatility-adjusted returns.

**Q-16. Solution: B**

If the intensity parameter (i.e., decay factor) is close to 1, there will be persistence (i.e., slow decay) in the estimate. The expression for the weight on day  $i$  has  $i$  in the exponent when it should be  $n$ . While a large sample size is generally preferred, some of the data may no longer be representative in a large sample.

**Q-17. Solution: C**

Each historical return (t) is replaced by:  $\text{return}(t) \times \text{volatility}(0)/\text{volatility}(t)$ . Then conduct typical historical simulation (HS) on adjusted return series

For example, if on the historical day (t), the  $\text{return}(t)$  was -2.0% and  $\text{volatility}(t)$  was 10%, while today's volatility estimate is 20%, then the adjusted return is  $-2.0\% \times 20\%/10\% = -4.0\%$ . In this way, "Actual returns in any period t are therefore increased (or decreased), depending on whether the current forecast of volatility is greater (or less than) the estimated volatility for period t. We now calculate the HS P/L using [the adjusted returns] instead of the original data set, and then proceed to estimate HS VaRs or ESs in the traditional way (i.e., with equal weights, etc.).

**Q-18. Solution: C**

Market-weighted historical simulation is not discussed in this topical. Age-weighted historical simulation weights observations higher when they appear closer to the event date. Volatility-weighted historical simulation adjusts for changing volatility levels in the data. Correlation-weighted historical simulation incorporates anticipated changes in correlation between assets in the portfolio.

**Q-19. Solution: B**

The volatility-weighting method adjusts historic returns for current volatility. Specifically, return at time t is multiplied by (current volatility estimate/ volatility estimate at time t). However, the actual procedure for calculating VaR using a historical simulation method is unchanged; it is only the inputted data that changes.

**Q-20. Solution: C**

The use of historical data in non-parametric analysis is a disadvantage, not an advantage. If the estimation period was quiet (volatile) then the estimated risk measures may understate (overstate) the current risk level. Generally, the largest VaR cannot exceed the largest loss in the historical period. On the other hand, the remaining choices are all considered advantages of non-parametric methods. For instance, the non-parametric nature of the analysis can accommodate skewed data, data points are readily available, and there is no requirement for estimates of covariance matrices.

**Q-21. Solution: C**

C is correct. The correlation-weighted approach is similar to the volatility-weighted approach in that both approaches adjust past returns that occurred under a different correlation or volatility regime to reflect the current correlation or volatility. In the correlation-weighted approach this is accomplished through matrix multiplication since correlations are represented with a matrix.

A is incorrect. This is a description of an equal-weighted approach, not an age-weighted approach.

B is incorrect. The volatility-weighted approach does adjust historical returns but does so in a manner that decreases the historical return if the historical volatility forecast was higher than the current volatility forecast, and increases the historical return if the historical volatility forecast was lower than the current volatility forecast. The volatility adjusted returns are given by the following formula:  $r^*_{t,l} = (\sigma_{T,l} / \sigma_{t,l}) r_{t,l}$  Where  $\sigma_{T,l}$  is the current volatility forecast,  $\sigma_{t,l}$  is the historical volatility forecast,  $r_{t,l}$  is the historical return, and  $r^*_{t,l}$  is the adjusted historical return.

D is incorrect. The filtered HS approach is not a simple filter but is a form of semi-parametric bootstrap model that adjusts returns within a conditional volatility framework.

**Q-22. Solution: C**

The POT approach generally has fewer parameters, but GEV approaches share the tail parameter  $\xi$ .

**Q-23. Solution: C**

There is a trade-off in setting the threshold. It must be high enough for the appropriate theorems to hold, but if set too high; there will not be enough observations to estimate the parameters.

**Q-24. Solution: B**

B is correct. As can be seen from the formula below, increasing  $u$  increases both VaR and ES even if  $n$  gets lower as  $u$  increases.

A is incorrect. Increasing the tail parameter value actually increases both VaR and ES.

C and D are incorrect. According to the peaks-over-threshold (POT) risk measure, the VaR and ES (in percentage) are computed as (note: the first equation is not necessary as the value of VaR is given):

$$\begin{aligned} \text{VaR} &= u + \left(\frac{\beta}{\xi}\right) \left\{ \left[ \frac{N}{n} (1 - \text{confidence level}) \right]^{-\xi} - 1 \right\} \\ &= 3 + \left(\frac{0.75}{0.22}\right) \left\{ \left[ \frac{740}{37} (1 - 0.99) \right]^{-0.22} - 1 \right\} = 4.45\% \end{aligned}$$

And

$$\text{ES} = \frac{4.45}{1 - 0.22} + \frac{0.75 - 0.22 \times 3}{1 - 0.22} = 5.82\%$$

**Q-25. Solution: A**

A key foundation of EVT is that as the threshold value is increased, the distribution of loss exceedances converges to a generalized Pareto distribution. Assuming the threshold is high enough, excess losses can be modeled using the generalized Pareto distribution. Thus, A is correct.

B is incorrect. If the tail parameter value of the generalized extreme-value (GEV) distribution goes to zero, and not infinity, then the distribution of the original data {not the GEV} could be a light-tail distribution such as normal or log-normal. In other words, the corresponding GEV distribution is a

Gumbel distribution.

C is incorrect. To apply EVT, the underlying loss distribution can be any of the commonly used distributions: normal, lognormal, t, etc.

D is incorrect. As the threshold value is decreased, the number of exceedances increases.

**Q-26. Solution: A**

A is correct. The regression method is the easiest method to apply, and the recovery of least squares estimates of the EV parameters from a regression is straightforward. The first step in applying the regression method is to order the sample of extreme values from lowest to highest.

B is incorrect. The use of an arbitrary (not random) number would correctly describe the Hill estimator, which is used in semi-parametric estimation method. However, this would still be an incorrect answer to this specific question.

C is incorrect. The main challenge with the semi-parametric method (not the moment-based method) is choosing the number of observations to include in the tail, as the parameter estimates can be sensitive to this number.

D is incorrect. The most popular semi-parametric estimation method is the Hill estimator, which is known to be consistent and asymptotically normal.

**Q-27. Solution: C**

Backtesting VaR models with higher confidence levels is difficult because the number of exceptions is not high enough to provide meaningful information.

**Q-28. Solution: B**

VaR models are based on static portfolios, while actual portfolios are constantly changing, and actual returns are complicated by intraday changes as well as profit and loss factors that result from commissions, fees, interest income, and bid-ask spreads. A risk manager can minimize these effects by backtesting with a relatively short time horizon such as a daily holding period.

**Q-29. Solution: C**

$T \text{ value} = (25 - 1000 \times 1\%) / \sqrt{1000 \times 1\% \times 99\%} = 4.74$

we reject the null hypothesis and decide that the model is a bad model, which implies a risk of type I error.

**Q-30. Solution: D**

Null hypothesis is  $H_0$ : Model is good with  $E[\text{exceptions}] = (1 - 95\%) \times 250 = 12.5$  exceptions

The standard error (standard deviation) of the binomial variable

$$= \sqrt{p(1-p)T} = \sqrt{5\% \times (1 - 5\%) \times 250} = 3.446$$

The test statistic is  $[19 - 12.5] \div 3.446 = 1.89$

In words, we observed 6.5 more exceptions (19 - 12.5) than expected if the model is good, which is 1.89 standard deviations away from the expected number of exceptions. Since we know that a 95% one-tailed normal confidence interval implies a 1.645 cutoff, we know that 1.645 is also the cutoff for a 90% two-tailed since the normal is symmetrical, this falls outside the acceptance region. We reject the null, assuming that luck does not explain this, and find the model faulty.

**Q-31. Solution: A**

A is correct. The concept tested here is the understanding of the difference between the VaR parameter for confidence (here, namely 95% vs. 99%) and the validation procedure confidence level (namely 95%), and how they interact with one another. Using a 95% VaR confidence level creates a narrower nonrejection region than using a 99% VaR confidence level by allowing a greater number of exceptions to be generated. This in turn increases the power of the backtesting process and makes for a more reliable test than using a 99% confidence level.

**Q-32. Solution: C**

In the case of a bank that changed positions more frequently during the day, a penalty should be considered, but it is not necessarily given. In the case of bad luck, no penalty is given, as would be the case for a bank affected by unpredictable movements in rates or markets. However, when risk models are not precise enough, a penalty is typically given since model accuracy could have easily been improved.

**Q-33. Solution: B**

The risk manager will reject the hypothesis that the model is correctly calibrated if the number  $x$  of losses exceeding the VaR is such that:

$$\frac{x - p \times T}{\sqrt{p(1 - p)T}} > z = 1.96$$

where  $p$  represents the failure rate and is equal to  $1 - 99\%$ , or  $1\%$ ; and  $T$  is the number of observations = 225. And  $z = 1.96$  is the two-tail confidence level quantile. If:

$$\frac{x - 0.01 \times 225}{\sqrt{0.01 \times (1 - 0.01) \times 225}} > z = 1.96$$

Then,  $x = 5.18$ . So the maximum number of exceedances would be 5 to conclude that the model is calibrated correctly..

**Q-34. Solution: A**

A is correct. Ideally, the analyst will use both the actual and hypothetical returns for backtesting

since both yield informative comparisons. If the model passes backtesting with hypothetical but not actual returns, the problem in the model lies with intraday trading. If the model does not pass backtesting with hypothetical returns, the modelling methodology should be re-examined.

B is incorrect. The descriptions of the cleaned returns are not correct, and these are not the correct two sets of returns that the analyst should use. Cleaned returns will also subtract any fees, commissions, net interest margin from actual returns. The correct definition of cleaned returns can sometimes be used as an approximation of hypothetical returns, but not the incorrectly defined returns stated here.

C is incorrect. These are not the correct two sets of returns that the analyst should use, and the description of cleaned returns is incorrect.

D is incorrect. There is not a return set that is called trading returns.

**Q-35. Solution: B**

B is correct. In the case of bad luck, no penalty is given, as would be the case for a bank affected by unpredictable movements in rates or markets. However, when risk models are not precise enough, a penalty is typically given since model accuracy could have easily been improved.

**Q-36. Solution: C**

C is correct. A model that produces too many exceptions may cause too little capital to be allocated to risk-taking units.

A is incorrect. Observing too many or too few exceptions is not related to the confidence level of the model.

B is incorrect. A model that produces too many exceptions is underestimating the market risk the bank faces.

D is incorrect. The bank's VaR model may see too many exceptions due to a poorly designed model, deliberately underestimating risk, or bad luck. The regulator may impose penalties to prevent the first two reasons, but the regulator may not impose penalties if the exceedances are the result of bad luck, or intraday trading.

**Q-37. Solution: A**

A is correct. As the sample size increases, the confidence region of the failure rate (expressed as  $N/T$ ) for a correctly specified model will shrink, and an incorrectly specified model should be able to be rejected more easily.

B is incorrect. A test with a low type I error rate and a low type II error rate is said to be powerful. The power of a test relates to the error rates of both type I and type II errors.

C is incorrect. A type I error is rejecting a correctly specified model.

D is incorrect. A type II error is not rejecting an incorrectly specified model.

**Q-38. Solution: B**

A is incorrect. It corresponds to a standard uniformly distributed PIT series.

C is incorrect. The independence property requires that each observed exception is not informative of future exceptions. Furthermore, iid standard uniformly distributed PIT series corresponds to the satisfaction of both unconditional coverage and independence property.

D is incorrect as per explained in C.

**Q-39. Solution: A**

A is correct. The bank derives the PITs by collecting the daily realization of the portfolio Profit/Loss and calculating the risk model's probability of observing a realization below the actual Profit/Loss.

**Q-40. Solution: D**

D is correct. A model is aggressively misspecified when the model distribution is too narrow, and we often see realized Profit/Loss in the tails of the distribution. In contrast, a model is conservatively misspecified when the model distribution is too wide, and the observed Profit/Loss clusters in the middle of the distribution.

**Q-41. Solution: D**

A is incorrect. Kolmogorov-Smirnov test is lack of sensitivity at the extremes of the distribution. It is effective in detecting differences in central masses of the two distributions.

B is incorrect. The Cramer-von Mises test is a variation of the Kolmogorov-Smirnov test that relies on the mean squared deviation of the distribution.

C is incorrect. In the context of VaR backtesting, the null hypothesis of all three goodness-of-fit tests is that the empirical CDF is cumulative standard uniform.

**Q-42. Solution: C**

A is incorrect. For VaR models, the basic purpose is to help the firm manage the risk of its positions.

B is incorrect. If the VaR is calculated based on the actual Profit/Loss history, the change in positions does not result in a change in VaR since the actual Profit/Loss history is unaffected by the change in positions.

D is incorrect. Beyond the consideration of how the model will be used, the conceptual soundness of the model can be considered using more quantitative tests such as the sensitivity analysis.

**Q-43. Solution: C**

A is incorrect. The sensitivity of VaR to the position is dependent on the share of the position in the portfolio and the sensitivity of the position value to the overall portfolio value.

B is incorrect. Data scarcity is not negligible. The approach described in this chapter requires that

you have a full time series of the change in value of the position. However, financial institutions often conduct the sensitivity analysis because the valuation data is missing or scarce.

D is incorrect. The portfolio composition of a large trading organization typically changes rapidly, which forms a challenge of performing the sensitivity analysis.

**Q-44. Solution: D**

A is incorrect. It may be desirable to base the confidence interval on more than the single point of the quantile estimate and instead estimate the confidence interval based on a calculation of the variance of the distribution of pseudo returns.

B is incorrect. The order statistics approach reorders portfolio value changes from lowest to highest.

C is incorrect. The order statistics approach also assumes independence of the changes in portfolio value over time.

**Q-45. Solution: B**

A is incorrect. There is rarely anything useful to say about the models from these plots other than one model seems more conservative than the other or that they seem to behave about the same.

C is incorrect. Berkowitz and O'Brien (2002) overcome the issue that banks rarely have two VaR models available to compare by estimating a GARCH (1, 1) VaR model on the Profit/Loss from trading by the bank.

D is incorrect. Rejecting the null would indicate that one model is a significantly better model under the loss function considered.

**Q-46. Solution: A**

The formula for systematic risk and specific risk in variance are:

$$\text{Systematic Risk} = \sum_{i=1}^n w_i^2 \times \beta_i^2 \times \sigma_m^2$$

$$\text{Specific Risk} = \sigma_p^2 - \text{Systematic Risk}$$

Using the given data, total systematic risk = 0.0028512 + 0.000729 + 0.0036 = 0.0071802

Portfolio Specific Risk = 0.0205 – 0.0071802 = 0.0168198 ≈ 0.0168

**Q-47. Solution: C**

Standard positions in the six-month zero:

$$\frac{30,000}{1 + \frac{2\%}{2}} = 29,703$$

Standard positions in the twelve-month zero:

$$\frac{1,000,000 + 30,000}{1 + 2.5\%} = 1,004,878.049$$

**Q-48. Solution: D**

General risk factor in market risk can be classied in four categories:

1. Equity price risk
2. Interest rate risk
3. Exchange rate risk
4. Commodity price risk

**Q-49. Solution: A**

Delta-normal VaR methods will provide accurate estimates of VaR for options only over those ranges in which the deltas of the contracts are stable. Deltas are normally unstable near the money and close to expiration.

**Q-50. Solution: B**

The diversified VaR is lower due to two factors. First, risk measures are not perfectly linear with maturity. Second, correlations are below unity, which reduces risk even further.

**Q-51. Solution: A**

With principal mapping, one risk factor is chosen that corresponds to the average portfolio maturity. With duration mapping, one risk factor is chosen that corresponds to the portfolio duration. With cash flow mapping, the portfolio cash flows are grouped into maturity buckets. Convexity mapping is not a method of VaR mapping for fixed income portfolios.

**Q-52. Solution: D**

D is correct. With duration mapping, a portfolio is replaced by a zero-coupon bond with maturity equal to the duration of the portfolio.

A is incorrect. Cash-flow mapping considers the present values of the cash flows placed to correspond to the maturities for which volatilities are provided. So, in cash-flow mapping, cash flows are grouped into maturity brackets.

B is incorrect. Cash-flow mapping considers the present values of the cash flows and uses the appropriate zero-coupon rate as the discount factor.

C is incorrect. Principal mapping is a simple method that considers the timing of redemption payments only. Correlations among zero-coupon bonds with different maturities are considered in cash-flow mapping.

**Q-53. Solution: C**

Under the cash flow mapping approach, each payment (and not only the last one) is associated with a different risk factor, so statement I. is incorrect. Statement II. is incorrect because the CF mapping approach is more correct than duration or maturity mapping.

**Q-54. Solution: C**

Mapping government bonds paying regular coupons onto zero coupon government bonds is an adequate process, because both categories of bonds are government issued and therefore have a very similar sensitivity to risk factors. However, this is not a perfect mapping since the sensitivity of both classes of bonds to specific risk factors (i.e., changes in interest rates) may differ.

**Q-55. Solution: B**

B is correct. With duration mapping, a portfolio is replaced by a zero-coupon bond with maturity equal to the duration of the portfolio. The risk of the hypothetical zeros is less than the risk of a coupon bond of comparable maturity. Therefore, the portfolio VaR using duration mapping is less than the portfolio VaR using principal mapping. Option A is not correct. Option C is incorrect because of the reason stated for the correctness of Option B. Option D is incorrect because of the same reason for the correctness of Option A.

**Q-56. Solution: B**

B is correct. The process used in the duration mapping approach is to replace the portfolio of fixed-income investments with a single zero-coupon with a maturity equal to the duration of the portfolio.

A is incorrect. Principal mapping overstates the true risk of the portfolio because it ignores coupon payments. Taking these payments into account decreases the duration of the portfolio, which decreases portfolio VaR. Therefore, principal mapping overstates the risk of the portfolio.

C is incorrect. Undiversified VaR and the inclusion of correlations are associated with the cash-flow mapping approach rather than the duration mapping approach. There is a difference between the VaR estimated using the principal mapping approach and the diversified VaR estimated using the duration mapping approach due to the intervening cash flows. The principal mapping approach ignores intervening cash flows while they are implicitly accounted for in the duration mapping approach.

D is incorrect. The VaR estimated using the cash-flow mapping approach is more accurate than the VaR estimated using the duration mapping approach since the cash-flow approach incorporates correlations into the estimation process.

**Q-57. Solution: C**

C is correct. We need to map the portfolio to a position in the underlying stock TUV. A deep in-the-

money call has a delta of approximately 1, a deep out-of-the-money call has a delta of approximately zero and forwards have a delta of 1.

The net portfolio has a delta ( $D_p$ ) of about  $1 \times 5,000 + 0 \times 20,000 + 1 \times 10,000 = 15,000$  and is approximately gamma neutral.

Let:

$Z = 2.326$  (99% confidence level)

$S$  = price per share of stock TUV = USD 52

$D_p$  = delta of the position = 15,000

$\sigma$  = volatility of TUV = 0.12

Therefore, the 1-day VaR estimate at 99% confidence level is computed as follows:

$$\alpha \cdot S \cdot D_p \cdot \sigma \cdot \sqrt{1/T} = (2.326) \cdot (52) \cdot (15,000) \cdot (0.12 / \sqrt{252}) = \text{USD } 13,714.67$$

**Q-58. Solution: B**

- I. If the correlation will rise in the future, enter correlation swaps as a party to pay fixed correlation will gain a profit. Paying a fixed rate in a correlation swap is also called "buying correlation".
- II. Another way of buying correlation (i.e., benefiting from an increase in correlation) is to buy put options on an index such as the Dow Jones Industrial Average (Dow) and sell put options on individual stock of the Dow.
- III. A further way to buy correlation is to pay fixed in a variance swap on an index and to receive fixed in variance swaps on individual components of the index. As a consequence, the present value for the variance swap buyer, the fixed variance swap payer, will increase. This increase is expected to outperform the potential losses from the short variance swap positions on the individual components.

**Q-59. Solution: A**

"I" is true. Using the copula approach, we can calculate the structures of correlation between variables separately from the marginal distributions. "IV" is also true. Correlation is a good measure of dependence when the measured variables are distributed as multivariate elliptical.

"II" is false. The correlation between transformed variables will not always be the same as the correlation between those same variables before transformation. Data transformation can sometimes alter the correlation estimate. "III" is also false. Correlation is not defined unless variances are finite.

**Q-60. Solution: B**

There is a -4% difference from the long-run mean correlation and April 2014 correlation ( $32\% - 36\% = -4\%$ ). The inverse of the  $\beta$  coefficient in the regression relationship implies a mean

reversion rate of 75%. Thus, the expected correlation for May 2014 is 33.0%:

$$S_t = a(\mu_s - S_{t-1}) + S_{t-1}$$

$$S_t = 0.75(32\% - 36\%) + 0.36 = 0.33$$

**Q-61. Solution: A**

The autocorrelation for a one-period lag is 23% for the same sample. The sum of the mean reversion rate (77% given the beta coefficient of -0.77) and the one-period autocorrelation rate will always equal 100%.

**Q-62. Solution: B**

The fixed CDS spread is valued based on the default probability of the reference asset (Canadian bond) and the joint default correlation of First Bank (counterparty) and Canada. A paper loss occurs for the investor (CDS spread buyer) if the correlation risk between First Bank and Canada increases because the value of the CDS spread will decrease. The investor has wrong-way risk if there is positive correlation risk between First Bank and Canada. Increasing correlation risk increases the probability that the worst case scenario occurs where both First Bank and Canada default.

**Q-63. Solution: D**

Correlation estimates tend to be very volatile when short term time horizons are considered.

**Q-64. Solution: D**

D is correct. The required probability is:

$$P\{t_{BB} \leq 2 \cap t_B \leq 2\}$$

$$= P\{[N^{-1}(Q_{BB}(t_{BB}))] \leq N^{-1}(Q_{BB}(2))] \cap [N^{-1}(Q_B(t_B)) \leq N^{-1}(Q_B(2))]\}$$

$$= P\{[X_{BB} \leq -1.209] \cap [X_B \leq -0.533]\}$$

A and B are both incorrect. In fact, under copula model, both  $x_{BB}$  and  $x_B$  are continuous random variables and thus both A and B correspond to zero probability. C is also incorrect because the transformation  $N^{-1}$  is not properly considered in this option.

**Q-65. Solution: A**

A is correct. Wrong-way risk arises when there is a positive default correlation between the reference asset and the CDS counterparty.

B is incorrect. The lower the correlation between the assets in a portfolio, the higher the return/risk (risk-adjusted returns) ratio.

C is incorrect. Gaussian copulas are used to measure the static default correlation risk of CDOs, they would not be used in a portfolio of pairs trades.

D is incorrect. Periods of systemic crises have the highest correlation risk as the change in correlation is often highest in these crises as correlations move closer to 1.

**Q-66. Solution: D**

$$\rho_{\text{realized}} = \frac{2}{n^2 - n} \sum_{i>j} \rho_{i,j} = \frac{2}{12} \times (0.1 + 0.2 + 0.2 + 0.2 + 0.3 + 0.4) \\ = 0.233333$$

The payoff of correlation swap buyer =  $0.233333 \times 1 \text{ million} = \$233,333$

**Q-67. Solution: B**

Correlation levels are lowest in strong economic growth times. In recessions, correlation levels typically increase.

Correlation volatility is lowest in an economic expansion and highest in normal economic states.

**Q-68. Solution: B**

The trader would need to adjust hedge as follows:

$$\$89.8 \text{ million} \times 1.0274 = \$92.26 \text{ million}$$

Thus, the trader needs to purchase additional TIPS worth \$2.46 million.

**Q-69. Solution: B**

Defining  $F^R$  and as  $F^N$  the face amounts of the real and nominal bonds, respectively, and their corresponding DV01 as  $DV01^R$  and  $DV01^N$ , a DV01 hedge is adjusted by the hedge adjustment factor, or beta, as follow:

$$F^R = F^N \times \frac{DV01^N}{DV01^R} \times \widehat{\beta} = 100,000 \times 0.072 / 0.051 \times 1.2 = 169,412$$

**Q-70. Solution: B**

Defining  $F^R$  and as  $F^N$  the face amounts of the real and nominal bonds, respectively, and their corresponding DV01s as  $DV01^R$  and  $DV01^N$ , a DV01 hedge is adjusted by the hedge adjustment factor, or beta, as follow:

$$F^R = F^N \times \frac{DV01^N}{DV01^R} \times \widehat{\beta} = 100M \times 0.065 / 0.088 \times 1.03 = 76.08 \text{ million}$$

This regression hedge approach suggests that for every \$100 million sold in T-bonds, Ryan should buy \$76.08 million in TIPS.

**Q-71. Solution: C**

A and B are incorrect because the statement is clearly incorrect.

D is incorrect. It is first noted that we can scale up the reverse regression hedge to yield the same

volatility as the original hedged position (i.e., hedging \$100m corporate bond with \$93.6m US Treasury). Taking the class example as an illustration: under the original hedged position, the risk is

$$\left| 100\text{m} \times \frac{0.2124}{100} \right| \times 0.0002 = \$425,000$$

while the risk under the reverse regression is

$$\left| 93.6\text{m} \times \frac{0.1910}{100} \right| \times 0.000209 = \$374,000$$

Of course, you can scale up the face value of the Treasury so that

$$\left| X \times \frac{0.1910}{100} \right| \times 0.000209 = \$425,000$$

$$\rightarrow X = 106.37\text{m}$$

Together with a beta of 0.921 in reverse regression, we hedge this position with \$88m corporate bonds. However, because this transaction does not hold \$100m in corporate bonds, the trades are completely different.

**Q-72. Solution: A**

This is the same underlying bond and interest rate tree as in the call option example from this topic. However, here we are valuing a put option.

The option value in the upper node at the end of year 1 is computed as:

$$\frac{2.44 \times 0.6 + 0.38 \times 0.4}{1.0599} = 1.52$$

The option value in the lower node at the end of year 1 is computed as:

$$\frac{0.38 \times 0.6 + 0 \times 0.4}{1.0444} = 0.22$$

The option value today is computed as:

$$\frac{1.52 \times 0.76 + 0.22 \times 0.24}{1.0300} = 1.17$$

**Q-73. Solution: C**

$$\left( \frac{1}{1+7\%} \times 50\% + \frac{1}{1+5\%} \times 50\% \right) / (1+6\%) = 0.89007$$

**Q-74. Solution: B**

The 3-year spot rate can be solved for using the following equation:

$$\frac{\$1}{(1+6\%)(1+8\%)(1+10\%)} = \frac{\$1}{(1+r)^3}$$

$$r = 7.988\%$$

**Q-75. Solution: A**

The left-hand side of Jensen's inequality is the expected price in one year using the 1-year spot rates of 8% and 4%.

$$E\left(\frac{\$1}{1+r}\right) = 0.5 \times \frac{\$1}{1+8\%} + 0.5 \times \frac{\$1}{1+4\%} = 0.94373$$

The expected price in one year using an expected rate of 6% computes the right-hand side of the inequality as:

$$\frac{\$1}{0.5 \times (1+8\%) + 0.5 \times (1+4\%)} = \frac{\$1}{1.06} = 0.9434$$

Next, divide each side of the equation by 1.06 to discount 1-year zero-coupon bond price for one more year at 6%. The price of the 2-year zero-coupon bond equals \$0.89031 (calculated as  $0.94373/1.06$ ), which is greater than \$0.89000 (the price of a 2-year zero-coupon bond discounted for two years at the expected rate of 6%). Thus, Jensen's inequality reveal that  $\$0.89031 > \$0.89000$ .

**Q-76. Solution: D**

A is incorrect. The Ito's Lemma gives the return decomposition in terms of return due to the passage of time and return due to changes in the rate.

B is incorrect. The decreases in rate will increase bond return in proportion to duration.

C is incorrect. The return due to the passage of time is the instantaneous forward rate.

**Q-77. Solution: A**

A is correct. The bond's expected return over the 6-month is:

$$(r_0 + \lambda D)dt$$

with  $r_0 = 2\%$ ,  $\lambda = 0.0015$ ,  $D = 4.2$ , and  $dt = 0.5$  (noted that  $dt$  is measured in year). Thus:

$$E\left[\frac{dP}{P}\right] = (2\% + 0.0015 \times 4.2) \times 0.5 = 1.32\%$$

**Q-78. Solution: B**

Under Model 1, it is true that the middle node recombines to the same current node. But these are future short-term rates; they are not the term structure: the term structure is spot rates at all maturities. Models that take the initial term structure implied by market prices are called arbitrage-free models. A different approach, however, is to start with assumptions about the interest rate process and about the risk premium demanded by the market for bearing interest rate risk and then derive the risk-neutral process. Models of this sort do not necessarily match the initial term structure and are called equilibrium models.

**Q-79. Solution: D**

Negative short-term interest rates can arise in models for which the terminal distribution of interest rates follows a normal distribution. The existence of negative interest rates does not make much economic sense since market participants would generally not lend cash at negative interest rates when they can hold cash earn a zero return. One method that can be used to address the potential for negative interest rates when constructing interest rates trees is to set all negative interest rates to zero. This localizes the change in assumptions to points in the distribution corresponding to negative interest rates and preserves the original rate free for all other observations. In comparison, adjusting the risk neutral probabilities would alter the dynamics across the entire range of interest rates and therefore not be an optimal approach.

When a model displays the potential for negative short-term interest rates, it can still be a desirable model to use in certain situations, especially in cases where the valuation depends more on the average path of the interest rate, such as in valuing coupon bonds. Therefore, the potential for negative rates does not automatically rule out the use of the model.

**Q-80. Solution: B**

Only statement 2 is correct. The Vasicek model implies decreasing volatility and non-parallel shifts from changes in short-term rates. The Ho-Lee model is actually more general than Model 2 (the no drift and constant drift models are special cases of the Ho-Lee model).

**Q-81. Solution: B**

Model 1 has a no-drift assumption. Using this model, the change in the interest rate is predicted as:

$$dr = \sigma dW = 0.8\% \times (-0.5) = -0.4\%$$

Since the initial rate was 5% and  $dr = -0.40\%$ , the new spot rate in one month is:

$$5\% - 0.40\% = 4.60\%$$

**Q-82. Solution: B**

This short rate process has an annualized drift of 0.36%, so it requires the use of Model 2 (with constant drift). The change in the spot rate is computed as:

$$dr = \lambda dt + \sigma dW = 0.36\% \times \frac{1}{12} + 0.8\% \times (-0.5) = -0.37\%$$

Since the initial short-term rate was 5% and  $dr$  is  $-0.37\%$ , the new spot rate in one month is:

$$5\% - 0.37\% = 4.63\%$$

**Q-83. Solution: C**

$$\text{node } [2, 0] = 4\% + (1\% + 0.8\%) \times \frac{1}{12} - 2 \times 2\% \times \sqrt{\frac{1}{12}} = 2.995\%$$

**Q-84. Solution: D**

The Ho-Lee model with drift is very flexible, allowing the drift terms each period to vary. Hence, the cumulative effect is additive. In contrast, the lognormal model with drift allows the drift terms to vary, but the cumulative effect is multiplicative.

**Q-85. Solution: C**

C is correct. In the CIR model, the basis-point volatility of the short rate is not independent of the short rate as other simpler models assume. The annualized basis-point volatility equals.

**Q-86. Solution: C**

Because we are using the Ho-Lee model, the drift is additive from period to period. Therefore, the expected short-term interest rate at the end of the second period (i.e., middle node) is  $5.4\% + 0.25\% - 0.1\% = 5.55\%$ .

**Q-87. Solution: C**

Choice c is correct: the Ho-Lee model incorporates a premium to each rate change that can be different at each point in time.

**Q-88. Solution: D**

In the CIR model, the basis-point volatility of the short rate is not independent of the short rate as other simpler models assume. The annualized basis-point volatility equals  $\sigma\sqrt{r}$  and therefore increases as a function of the square root of the rate.

**Q-89. Solution: B**

Choice B is correct: the Vasicek model incorporates mean reversion. The flexibility of the model also allows for risk premium, which enters into the model as constant drift or a drift that changes over time.

**Q-90. Solution: C**

C is correct. Negative short-term interest rates can arise in models for which the terminal distribution of interest rates follow a normal distribution. The existence of negative interest rates (although possible) does not make much economic sense since market participants would generally not lend cash at negative interest rates when they can hold cash and earn a zero return. One method that can be used to address the potential for negative interest rates when constructing interest rate trees is to set all negative interest rates to zero. This localizes the change in assumptions to points in the distribution corresponding to negative interest rates and preserves

the original rate tree for all other observations. In comparison, adjusting the risk neutral probabilities would alter the dynamics across the entire range of interest rates and therefore not be an optimal approach.

When a model displays the potential for negative short-term interest rates, it can still be a desirable model to use in certain situations, especially in cases where the valuation depends more on the average path of the interest rate, such as in valuing coupon bonds. Therefore, the potential for negative rates does not automatically rule out the use of the model.

**Q-91. Solution: B**

B is correct. The Vasicek model incorporates mean reversion. The flexibility of the model also allows for risk premium, which enters into the model as constant drift or a drift that changes over time.

**Q-92. Solution: A**

A is correct. The impact of volatility to the change in the interest rate between date 1 and date 2 will be the same at any node on date 2. Since the standard deviation of  $dw$  is  $\sqrt{dt}$ , the standard deviation of the rate change is  $\sigma(t) * dw = \sigma(t) * \sqrt{dt}$ . So, the volatility component of the change in interest rate is  $0.0080\sqrt{1/12} = 0.0023$ , 23 bps, up or down.

**Q-93. Solution: C**

C is correct. In the CIR model, basis-point volatility is a function of the yield volatility, which is assumed to be constant and the square root of the short rate. Therefore, by assumption yield volatility is constant and basis-point volatility varies.

A is incorrect. CIR states that when short-term interest rates are very low, basis-point volatility will also be very low, which decreases the possibility of negative interest rates. This possibility can be removed entirely if a positive drift factor that is larger than the impact of volatility is included in the model.

B is incorrect. In CIR, basis-point volatility is specified as an increasing function of the short rate.

D is incorrect. Basis-point volatility and yield volatility are measured in different units, but they do not measure the same volatility in the CIR model.

**Q-94. Solution: A**

A is correct. The equation for determining the expected short-term interest rate after some number of years (T) when the interest rate process follows a Vasicek model is given as follows (equation 13.23 in the Market Risk book):

$$r_0 * e^{-kT} + \theta * (1 - e^{-kT})$$

Using the information provided, the expected short-term interest rate after 8 years is:

$$0.0335 * e^{-0.06*8} + 0.0455 * (1 - e^{-0.06*8}) = 0.0335 * 0.6188 + 0.0455 * 0.3812 = 0.0381$$

The half-life of the short-term interest rate is given as being equal to  $\ln(2)/k$  (see equation 13.25 in Market Risk book). Therefore, the half-life in this example is  $\ln(2)/0.06 = 11.55$  years.

B is incorrect. The half-life is incorrectly calculated as being  $1/k$ , or  $1/0.06$ , 16.7 years.

C is incorrect. The expected short-term interest rate is incorrectly estimated by switching the position of  $e^{-kT}$  and  $1 - e^{-kT}$ , which then yields 4.09%.

D is incorrect. Both errors made in B and C are made in this answer choice.

**Q-95. Solution: B**

A is incorrect. The long-term factor mean reverts to a constant with a constant volatility parameter.

C is incorrect. The evolution of medium-term factor and long-term factor is modelled with a pre-specified correlation parameter, while the volatility term in short-term rate process is dropped in order to fit the real-world situation.

D is incorrect. Under the Gauss+ model, the speed of mean reversion from short-term to medium-term is faster than that from medium-term to long-term.

**Q-96. Solution: B**

A is incorrect. The structure of Vasicek model is relatively simple. Together with a certain level of explanatory power on the term structure dynamics, the Vasicek model has some limited uses in practice.

C is incorrect. The Gauss+ model is well-known among practitioners as a tool for both proprietary trading and hedging.

D is incorrect. The Gauss+ model's concepts and assumptions are intuitively appealing, and it reasonably balances the goals of model tractability and of capturing the empirical complexity of the term structure dynamics.

**Q-97. Solution: B**

According to the Gauss+ model, the  $dr_t$  in 3-month is calculated as

$$\begin{aligned} dr_t &= \alpha_r(m_t - r_t)dt \\ &= 1.0547 \times (4.90\% - 2.25\%) \times \frac{3}{12} \\ &= 0.70\% \end{aligned}$$

The expected short-term rate in 3-month is therefore  $2.25\% + 0.70\% = 2.95\%$ .

**Q-98. Solution: C**

The half-life calculation under the Gauss+ model is the same as that under the Vasicek model

(because both of them adopt the Ornstein-Uhlenbeck process that models the mean reversion).

For the long-term factor, its half-life is calculated as

$$\begin{aligned}\tau &= \frac{\ln(2)}{\alpha_l} \\ &= \frac{\ln(2)}{0.0165} \\ &= 42.01\end{aligned}$$

**Q-99. Solution: C**

The long-term factor under the Gauss+ model follows the following process:

$$dl_t = \alpha_l(\mu - l_t)dt + \sigma_l dW_t^1$$

With  $dl_t = 0.40\%$ ,  $\alpha_l = 0.04$ ,  $l_t = 3.0\%$ ,  $\sigma_l = 120$  bps,  $dW_t^1 = 0.24$  and  $dt = 0.5$ , we have

$$\begin{aligned}0.40\% &= 0.04 \times (\mu - 3.0\%) \times 0.5 + 1.20\% \times 0.24 \\ &\rightarrow \mu = 8.60\%\end{aligned}$$

**Q-100. Solution: D**

A is incorrect. One should first regress the changes in the zero yields of various terms on the changes in the zero yields of medium- long-term bonds.

B is incorrect. We find  $\mu$  that minimizes the sum of the squared errors of observed yields relative to model yields across the whole data sample.

C is incorrect. While the model ultimately fits the two- and ten-year forward rates, the medium- and long-term factors are in no way those forward rates themselves.

**Q-101. Solution: A**

A volatility smile is a common graphical shape that results from plotting the strike price and implied volatility of a group of options with the same expiration date. Since the volatility smile is downward sloping to the right, the implied distribution has a fatter left tail compared to the lognormal distribution of returns. This means that an extreme decrease in the DAX has a higher probability of occurrence under the implied distribution than the lognormal. The ES will therefore be larger when the methodology is modified.

**Q-102. Solution: D**

A plot of the implied volatility of an option as a function of its strike price demonstrates a pattern known as the volatility smile or volatility skew. The implied volatility decreases as the strike price increases. Thus, all else equal, a risk monitoring system which assumes constant volatility for equity returns will understate the implied volatility for a long position in a deep-in-the-money call.

**Q-103. Solution: D**

There is higher implied price volatility for low strike price equity options. “Crashophobia” is based on the idea that large price declines are more likely than assumed in Black-Scholes-Merton prices, not that volatility increases when prices decline. Compared to the lognormal distribution, traders believe the probability of large down movements in price is higher than large up movements. Increasing leverage at lower equity prices suggests increasing volatility.

**Q-104. Solution: C**

The prices obtained with C are the right ones because they correspond to prices at which you could sell or buy the options.

**Q-105. Solution: A**

Based on the put-call parity,  $p_{BS} + S_0e^{-qT} = c_{BS} + Ke^{-rT}$  and  $p_{mkt} + S_0e^{-qT} = c_{mkt} + Ke^{-rT}$

We can know that:

$$p_{BS} - p_{mkt} = c_{BS} - c_{mkt}$$

And  $c_{BS} = \$3.50$ ,  $c_{mkt} = \$3.00$ ,  $p_{BS} = \$2.00$ .

So  $p_{mkt} = \$1.50$ .

Therefore A is the correct answer.

**Q-106. Solution: C**

Explanation: An empirical distribution with a fat right tail generates a higher implied volatility for higher strike prices due to the increased probability of observing high underlying asset prices.

**Q-107. Solution: D**

Away-from-the-money currency options have greater implied volatility than at-the-money currency options, this pattern is a volatility smile.

**Q-108. Solution: A**

Leverage changes and “crashophobia” have both been offered as explanations for the inverse relation between implied volatility and asset prices for equity options. At lower equity prices, the firm’s capital structure is more leveraged, resulting in greater price volatility.

**Q-109. Solution: A**

Currency options exhibit volatility smiles because the at-the-money options have lower implied volatility than away-from-the-money options.

Equity traders believe that the probability of large price decrease is greater than the probability of large price increase. Currency trades’ beliefs about volatility are more symmetric as there is no

large skew in the distribution of expected currency values (i.e. there is a greater chance of large price movements in either direction).

**Q-110. Solution: A**

A downward sloping volatility skew indicates that out of the money puts are more expensive than predicted by the Black-Scholes-Merton model and out of the money calls are cheaper than expected predicted by the Black-Scholes-Merton model. The implied distribution has fat left tails and thin right tails.

**Q-111. Solution: A**

The implied distribution of the underlying equity prices derived using the general volatility smile of equity options has a heavier left tail and a less heavy right tail than a lognormal distribution of underlying prices. Therefore, using the lognormal distribution of prices causes deep-out-of-the-money call options on the underlying to be priced relatively high.

The implied distribution of underlying foreign currency prices derived using the general volatility smile of foreign currency options has heavier tail than a lognormal distribution of underlying prices. Therefore, using the lognormal distribution of prices causes deep-out-of-the-money call options on the underlying to be priced relatively low.

**Q-112. Solution: B**

An in-the-money call has a strike price below 30. Therefore, using the chart above its implied volatility is greater than the at-the-money volatility, so using the at-the-money volatility would result in pricing an in-the-money call option lower than its fair price.

**Q-113. Solution: B**

B is correct. For a foreign currency option, the implied distribution gives a relatively high price for the option. The implied volatility is relatively low for at-the-money options, but it becomes higher as the option moves either in-the-money or out-of-the-money. Thus, the implied distribution has heavier tails than the lognormal distribution.

**Q-114. Solution: B**

B is correct. The implied distribution of the underlying equity prices derived using the general volatility smile of equity options has a heavier left tail and a less heavy right tail than a lognormal distribution of underlying prices. Therefore, using the implied distribution of prices causes deep-out-of-the-money call options on the underlying to be priced relatively low compared with using the lognormal distribution. The implied distribution of underlying foreign currency prices derived using the general volatility smile of foreign currency options has heavier tails than a lognormal

distribution of underlying prices .Therefore, using the implied distribution of prices causes deep-out-of-the-money call options on the underlying to be priced relatively high compared with using the lognormal distribution.

**Q-115. Solution: C**

C is correct. There are two interpolations to be made to find the correct reference implied volatility. One is time and the other is the K/S0 ratio.

To get 7-month reference implied volatilities with a K/S0 ratio of 1.075, interpolate between 6-month and 1-year in the K/S ratios of 1.05 and 1.10.

We need to add one month's worth of the difference between the 6-month and 1- year volatilities to the 6-month volatility:  $9.10 + 1/6*(9.55-9.10) = 9.175$ ;  $9.45 + 1/6*(9.75-9.45) = 9.5$ . Then average these two numbers since 1.075 is exactly between 1.05 and 1.1. Thus,  $(9.175+9.5)/2=9.3375$ .

A is incorrect. This uses the volatilities at the 1.05 K/S0 ratio and only interpolates time.

B is incorrect. This uses the volatilities at 6 months and only interpolates K/S0.

D is incorrect. This uses the volatilities at 12 month and only interpolates K/S0.

**Q-116. Solution: C**

C is correct. Demand for protective puts will increase the price of these puts, which will increase their implied volatilities, which creates a downward sloping volatility skew.

A is incorrect. The volatility smile commonly seen in equity options is downward sloping with higher implied volatility in lower strike prices and lower implied volatility in higher strike prices. This implies a greater probability of seeing a price in the left hand tail and a lower probability of seeing a price in the right hand tail than in the lognormal distribution.

B is incorrect. The volatility smile commonly seen in foreign exchange rate options implies a greater probability of the future rate being either <1 standard deviation or >2 standard deviations away from the mean, and a lower probability of the future rate being between 1 and 2 standard deviations away from the mean.

D is incorrect. Since the central bank announcements are unexpected the volatility curve will not be shaped with higher implied volatilities at-the-money than out-of-the- money. Protection will most likely be bought using out-of-the-money options which will create a volatility smile, rather than a frown.

**Q-117. Solution: C**

The 1996 amendment to Basel I and Basel II.5 use VaR with a 99% confidence interval and the FRTB uses the expected shortfall with a 97.5% confidence interval. Basel I market risk capital requirements produced a very current result because the 10-day horizon incorporated a recent period of time. The FRTB does not require adding a stressed VaR to the expected shortfall

calculation. It was Basel II.5 that required the addition of a stressed VaR.

**Q-118. Solution: B**

The two types of risk recognized by the incremental risk charge are: (1) credit spread risk, and (2) jump-to-default risk. Jump-to-default risk is measured by 99.9% VaR and not 97.5% expected shortfall.

**Q-119. Solution: D**

D is correct. The Basel committee has moved from the VaR and stressed VaR measures used in Basel I and Basel II.5 to the stressed ES measure used in FRTB.

A is incorrect. The reasoning is reversed. FRTB allows market risk to be calculated at the trading desk level.

B is incorrect. The FRTB is a culmination of Basel committee efforts to place less reliance on internal models approach. Under FRTB, all banks must calculate market risk capital using a standardized approach, even if they have been approved to use an internal models approach.

C is incorrect. The FRTB introduces 5 different liquidity horizons that are better matched to the liquidity horizons of different risk factors than the earlier 10-day horizons used in Basel I and Basel II.5.

**Q-120. Solution: B**

B is correct. If market liquidity decreases, the liquidity horizon (the time it takes to liquidate a position) will increase, and therefore, the credit risk faced by the bank will typically increase due to the longer amount of time the bank will be forced to hold at least some portion of the position.

A is incorrect. This answer choice incorrectly states the impact of market liquidity on the liquidity horizon. If market liquidity decreases, the liquidity horizon (the time it takes to liquidate a position) will not decrease but will increase, and therefore, the credit risk faced by the bank will typically increase.

C is incorrect. This answer choice incorrectly states the impact of liquidity horizon on the credit risk exposure. If market liquidity increases, the liquidity horizon (the time it takes to liquidate a position) will decrease, and therefore, the credit risk faced by the bank will typically decrease, not increase, due to the shorter amount of time the bank will be forced to hold at least some portion of the position.

D is incorrect. This answer choice incorrectly states the impact of market liquidity on the liquidity horizon. If market liquidity increases, the liquidity horizon (the time it takes to liquidate a position) will decrease, not increase, and therefore, the credit risk faced by the bank will typically decrease due to the shorter amount of time the bank will be forced to hold at least some portion of the position.